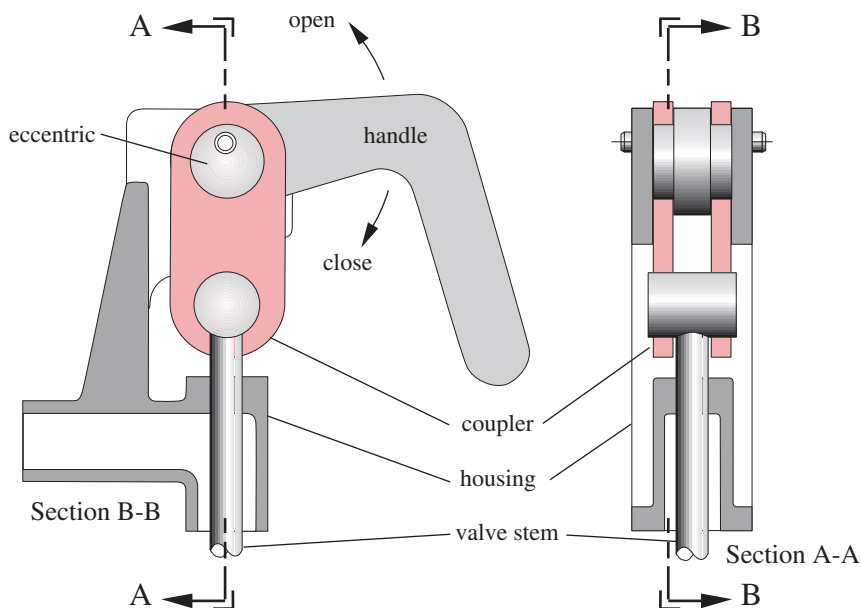
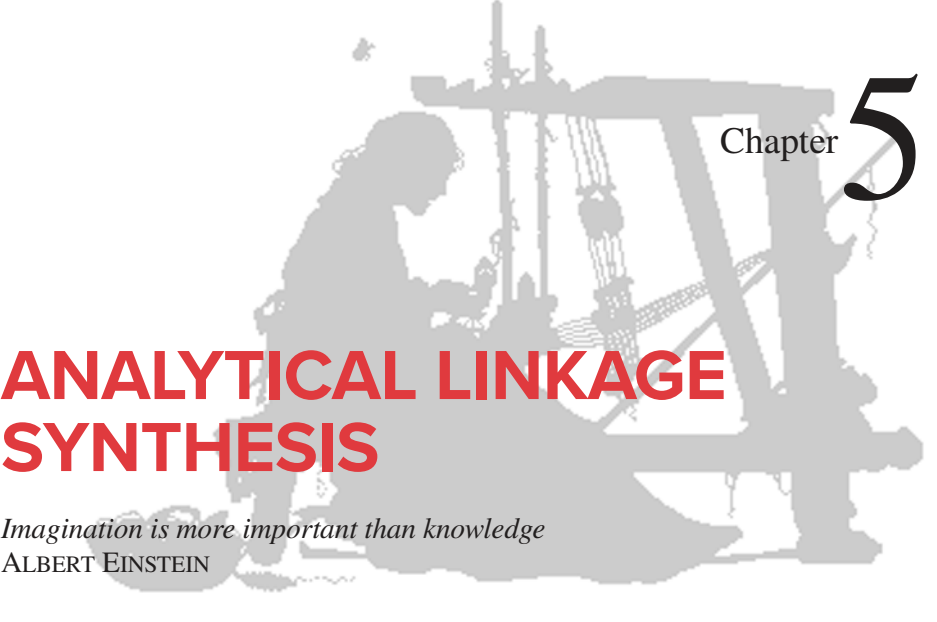


**FIGURE P4-20**

Problem 4-66

- 4-66 Figure P4-20 shows a cut-away view of a mechanism that opens and closes a remote valve by means of a long rod (valve stem) that moves up and down. The handle has two round bosses (eccentrics) whose centers are offset from the pivot by 6 mm. The eccentrics are connected to the valve stem by a coupler consisting of two identical links whose pivot holes have a center distance of 46 mm. It is an inline crank-slider mechanism. For the 180-degree-motion of the handle from closed to fully open, find and plot the stroke of the valve stem as a function of the angle of the handle.
- 4-67 For the linkage in Figure 3-32a, calculate and plot the angular displacement of links 3 and 4 and the path coordinates of point P with respect to the angle of the input crank O_2A for one revolution. The link lengths and coupler point data are: $L_1 = 3.72$, $L_2 = 1.00$, $L_3 = 1.94$, $L_4 = 3.72$, $p = 3.06$, and $\delta_3 = -20^\circ$.



Chapter 5

ANALYTICAL LINKAGE SYNTHESIS

Imagination is more important than knowledge
ALBERT EINSTEIN

5.0 INTRODUCTION *View the lecture video (48:17)*[†]

With the fundamentals of position analysis established, we can now use these techniques to **synthesize linkages** for specified output positions **analytically**. The synthesis techniques presented in Chapter 3 were strictly graphical and somewhat intuitive. The **analytical synthesis** procedure is algebraic rather than graphical and is less intuitive. However, its algebraic nature makes it quite suitable for computerization. These analytical synthesis methods were originated by Sandor^[1] and further developed by his students Erdman,^[2] Kaufman,^[3] and Loerch et al.^{[4], [5]}

[†] http://www.designofmachinery.com/DOM/Analytical_Linkage_Synthesis.mp4

5.1 TYPES OF KINEMATIC SYNTHESIS

Erdman and Sandor^[6] define three types of kinematic synthesis, **function**, **path**, and **motion generation**, which were discussed in Section 3.2. Brief definitions are repeated here for your convenience.

FUNCTION GENERATION is defined as *the correlation of an **input function** with an **output function** in a mechanism*. Typically, a double-rocker or crank-rocker is the result, with pure rotation input and pure rotation output. A slider-crank linkage can be a function generator as well, driven from either end, i.e., rotation in and translation out or vice versa.

PATH GENERATION is defined as *the control of a **point** in the plane such that it follows some prescribed path*. This is typically accomplished with a fourbar crank-rocker or double-rocker, wherein a point on the coupler traces the desired output path. No attempt is made in path generation to control the orientation of the link which contains the point of

interest. The coupler curve is made to pass through a set of desired output points. However, it is common for the timing of the arrival of the coupler point at particular locations along the path to be defined. This case is called *path generation with prescribed timing* and is analogous to function generation in that a particular output function is specified.

MOTION GENERATION is defined as *the control of a line in the plane such that it assumes some sequential set of prescribed positions*. Here orientation of the link containing the line is important. This is typically accomplished with a fourbar crank-rocker or double-rocker, wherein a point on the coupler traces the desired output path and the linkage also controls the angle of the coupler link containing the output line of interest.

5.2 TWO-POSITION SYNTHESIS FOR ROCKER OUTPUT

Example 3-1 showed a simple graphical technique for synthesis of a non-quick-return, Grashof fourbar linkage to drive a rocker through an angle. This technique was employed in later examples (e.g., 3-2, 3-4, 3-6) to construct a driver dyad to move a synthesized fourbar linkage through its desired range of motion, thus creating a Watt sixbar linkage. The rocker excursion cannot exceed 180° theoretically but should be limited to about 120° practically, which will give minimum transmission angles of 30° . The same dyad synthesis procedure can be done analytically and will prove to be of value in combination with the other synthesis techniques presented in this chapter.

Figure 5-0 shows the same problem as Figure 3-4 with generic annotation suitable for analytical determination of link lengths for the driver dyad. Link 4 (which might represent the input link to the next stage of the resulting Watt sixbar) is here the output link to be driven by a dyad consisting of links 2 and 3, whose lengths, along with that of the ground link 1 and its pivot location O_2 , are to be determined. The pivot location O_4 (defined in any convenient coordinate system XY), the initial angle θ_4 , and the excursion angle β are given. The procedure is as follows:*

* This procedure was provided by Prof. Pierre Laroche of South Dakota School of Mines.

First choose a suitable location on link 4 to attach link 3, here labeled B_1 and B_2 in its extreme locations. This defines R_4 , the length of link 4. These points can be defined in the chosen coordinate system as:

$$\begin{aligned} B_{1x} &= O_{4x} + R_4 \cos(\theta_4) & B_{1y} &= O_{4y} + R_4 \sin(\theta_4) \\ B_{2x} &= O_{4x} + R_4 \cos(\theta_4 + \beta) & B_{2y} &= O_{4y} + R_4 \sin(\theta_4 + \beta) \end{aligned} \quad (5.0a)$$

The vector $\mathbf{M}$ is the position difference between vectors $\mathbf{R}_{B_2}$ and $\mathbf{R}_{B_1}$

$$\mathbf{M} = \mathbf{R}_{B_2} - \mathbf{R}_{B_1} \quad (5.0b)$$

The parametric equation for line $\mathbf{L}$ can be written as:

$$\mathbf{L}(u) = \mathbf{R}_{B_1} + u\mathbf{M} \quad -\infty \leq u \leq \infty \quad (5.0c)$$

We want the resulting linkage to be a Class 1 Grashof crank rocker. We can achieve this by placing the crank pivot O_2 suitably far from B_1 along line $\mathbf{L}$. Let $M = |\mathbf{M}|$. It will be a Class 2 (non-Grashof) linkage when $B_1O_2 < M$, become Class 3 (Grashof with change points) when $B_1O_2 = M$, be Class 1 when $B_1O_2 > M$, and will again approach Class 3 when $B_1O_2 \gg M$. A reasonable range for B_1O_2 seems to be two to three times M .

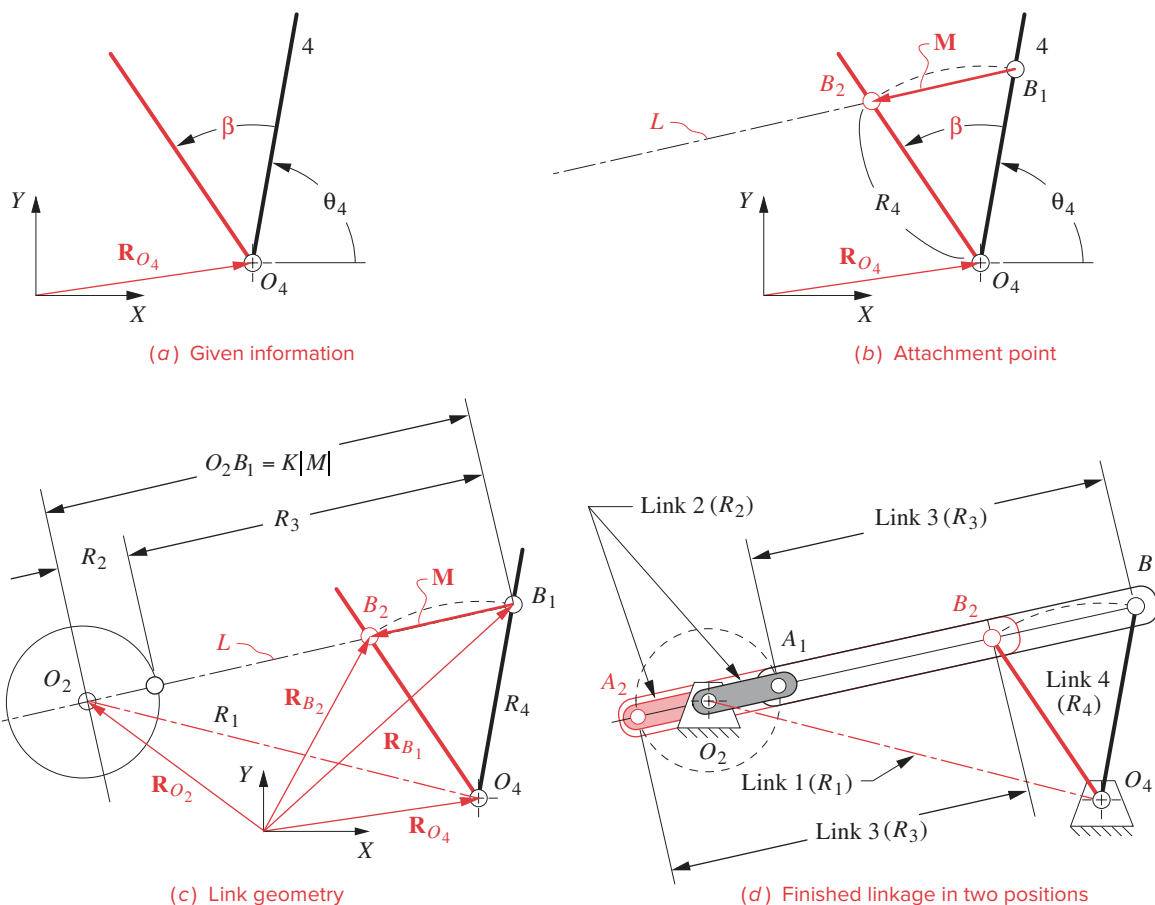


FIGURE 5-0

Analytical two-position synthesis with rocker output (non-quick-return)

$$\text{Let: } \mathbf{R}_{O_2} = \mathbf{R}_{B_1} \pm K\mathbf{M} \quad 2 < K < 3 \quad (5.0d)$$

As shown in Example 3-1, the length of the crank must be half the length of vector $\mathbf{M}$:

$$R_2 = 0.5|\mathbf{M}| = R_4 \sin(\beta/2) \quad (5.0e)$$

where β is in radians. Link 3 can be found by subtracting R_2 from the magnitude of $\mathbf{R}_{B_1} - \mathbf{R}_{O_2}$ and link 1 is found by subtracting $\mathbf{R}_{O_2}$ from $\mathbf{R}_{O_4}$.

$$R_3 = |\mathbf{R}_{B_1} - \mathbf{R}_{O_2}| - R_2; \quad R_1 = |\mathbf{R}_{O_4} - \mathbf{R}_{O_2}| \quad (5.0f)$$

This algorithm will result in a Grashof crank rocker mechanism that drives the rocker through the specified angle with no quick return.

5.3 PRECISION POINTS

The *points, or positions, prescribed for successive locations of the output (coupler or rocker) link in the plane* are generally referred to as **precision points** or **precision positions**. The number of precision points which can be synthesized is limited by the number of equations available for solution. The fourbar linkage can be synthesized by closed-form methods for up to five precision points for motion or path generation with prescribed timing (coupler output) and up to seven points for function generation (rocker output). Synthesis for two or three precision points is relatively straightforward, and each of these cases can be reduced to a system of linear simultaneous equations easily solved on a calculator. The four or more position synthesis problems involve the solution of nonlinear, simultaneous equation systems, and so are more complicated to solve, requiring a computer.

Note that these analytical synthesis procedures provide a solution which will be able to “be at” the specified precision points, but no guarantee is provided regarding the linkage’s behavior between those precision points. It is possible that the resulting linkage will be incapable of moving from one precision point to another due to the presence of a toggle position or other constraint. This situation is actually no different than that of the graphical synthesis cases in Chapter 3, wherein there was also the possibility of a toggle position between design points. In fact, these analytical synthesis methods are just an alternate way to solve the same multiposition synthesis problems. One should still build a simple cardboard model of the synthesized linkage to observe its behavior and check for the presence of problems, even if the synthesis was performed by an esoteric analytical method.

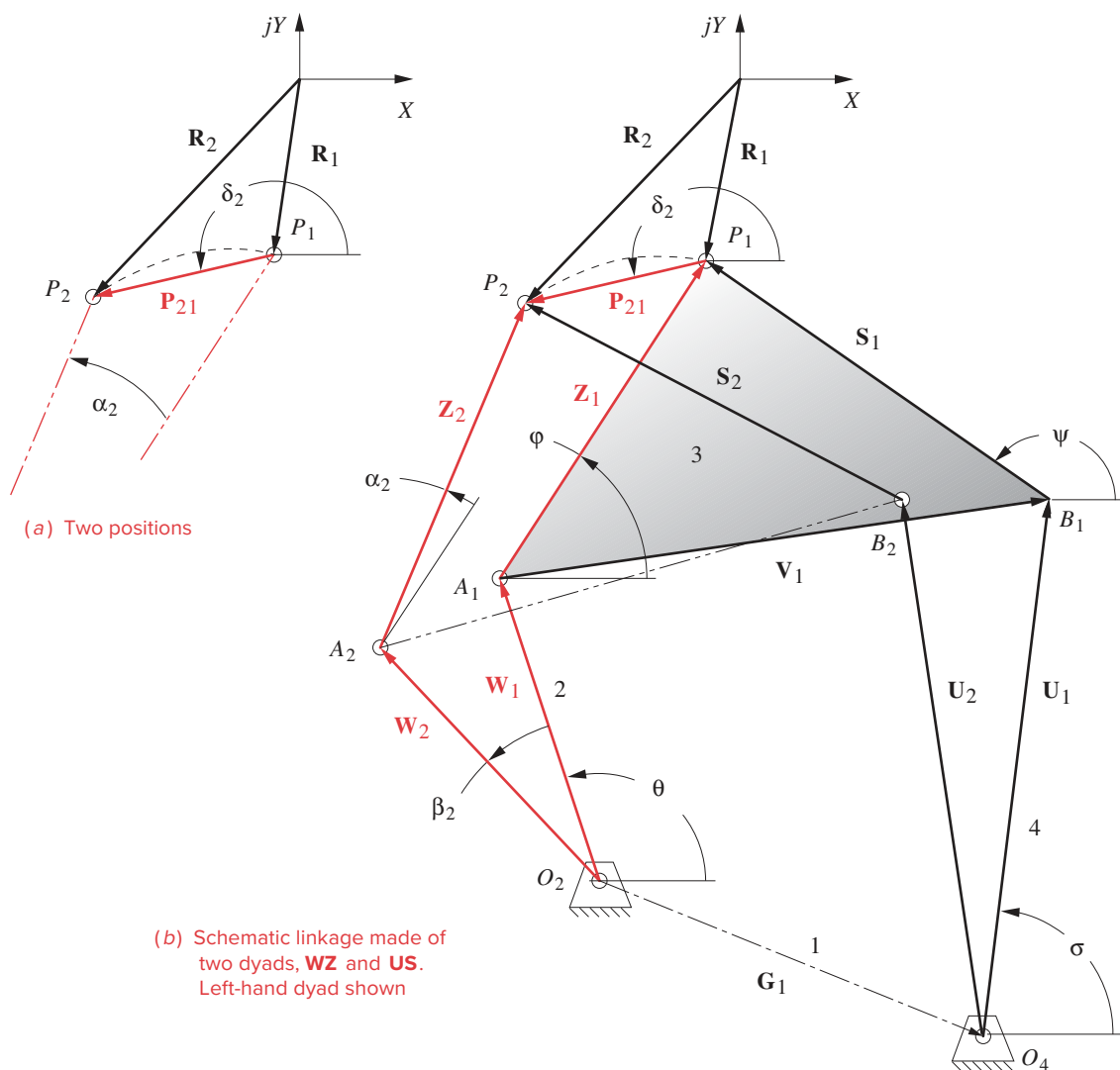
5.4 TWO-POSITION MOTION GENERATION BY ANALYTICAL SYNTHESIS

Figure 5-1 shows a fourbar linkage in one position with a coupler point located at a first precision position P_1 . It also indicates a second precision position (point P_2) to be achieved by the rotation of the input rocker, link 2, through an as yet unspecified angle β_2 . Note also that the angle of the coupler link 3 at each of the precision positions is defined by the angles of the position vectors $\mathbf{Z}_1$ and $\mathbf{Z}_2$. The angle ϕ corresponds to the angle θ_3 of link 3 in its first position. This angle is unknown at the start of the synthesis and will be found. The angle α_2 represents the angular change of link 3 from position one to position two. This angle is defined in the problem statement.

It is important to realize that the linkage as shown in the figure is schematic. Its dimensions are unknown at the outset and are to be found by this synthesis technique. Thus, for example, the length of the position vector $\mathbf{Z}_1$ as shown is not indicative of the final length of that edge of link 3, nor are the lengths (W, Z, U, V) or angles ($\theta, \phi, \sigma, \psi$) of any of the links as shown predictive of the final result.

The problem statement is:

Design a fourbar linkage which will move a line on its coupler link such that a point P on that line will be first at P_1 and later at P_2 and will also rotate the line through an angle α_2 between those two precision positions. Find the lengths and angles of the four links and the coupler link dimensions A_1P_1 and B_1P_1 as shown in Figure 5-1.

**FIGURE 5-1**

Two-position analytical synthesis

The two-position analytical motion synthesis procedure is as follows:

Define the two desired precision positions in the plane with respect to an arbitrarily chosen global coordinate system XY using position vectors $\mathbf{R}_1$ and $\mathbf{R}_2$ as shown in Figure 5-1a. The change in angle α_2 of vector $\mathbf{Z}$ is the rotation required of the coupler link. Note that the position difference vector $\mathbf{P}_{21}$ defines the displacement of the output motion of point P and is defined as:

$$\mathbf{P}_{21} = \mathbf{R}_2 - \mathbf{R}_1 \quad (5.1)$$

The dyad $\mathbf{W}_1\mathbf{Z}_1$ defines the left half of the linkage. The dyad $\mathbf{U}_1\mathbf{S}_1$ defines the right half of the linkage. Note that $\mathbf{Z}_1$ and $\mathbf{S}_1$ are both embedded in the rigid coupler (link 3), and both of these vectors will undergo the same rotation through angle α_2 from position 1 to position 2. The pin-to-pin length and angle of link 3 (vector $\mathbf{V}_1$) is defined in terms of vectors $\mathbf{Z}_1$ and $\mathbf{S}_1$.

$$\mathbf{V}_1 = \mathbf{Z}_1 - \mathbf{S}_1 \quad (5.2a)$$

The ground link 1 is also definable in terms of the two dyads.

$$\mathbf{G}_1 = \mathbf{W}_1 + \mathbf{V}_1 - \mathbf{U}_1 \quad (5.2b)$$

Thus if we can define the two dyads $\mathbf{W}_1, \mathbf{Z}_1$, and $\mathbf{U}_1, \mathbf{S}_1$, we will have defined a linkage that meets the problem specifications.

We will first solve for the left side of the linkage (vectors $\mathbf{W}_1$ and $\mathbf{Z}_1$) and later use the same procedure to solve for the right side (vectors $\mathbf{U}_1$ and $\mathbf{S}_1$). To solve for $\mathbf{W}_1$ and $\mathbf{Z}_1$, we need only write a vector loop equation around the loop which includes both positions P_1 and P_2 for the left-side dyad. We will go clockwise around the loop, starting with $\mathbf{W}_2$.

$$\mathbf{W}_2 + \mathbf{Z}_2 - \mathbf{P}_{21} - \mathbf{Z}_1 - \mathbf{W}_1 = 0 \quad (5.3)$$

Now substitute the complex number equivalents for the vectors.

$$we^{j(\theta+\beta_2)} + ze^{j(\phi+\alpha_2)} - p_{21}e^{j\delta_2} - ze^{j\phi} - we^{j\theta} = 0 \quad (5.4)$$

The sums of angles in the exponents can be rewritten as products of terms.

$$we^{j\theta}e^{j\beta_2} + ze^{j\phi}e^{j\alpha_2} - p_{21}e^{j\delta_2} - ze^{j\phi} - we^{j\theta} = 0 \quad (5.5a)$$

Simplifying and rearranging:

$$we^{j\theta}(e^{j\beta_2} - 1) + ze^{j\phi}(e^{j\alpha_2} - 1) = p_{21}e^{j\delta_2} \quad (5.5b)$$

Note that the lengths of vectors $\mathbf{W}_1$ and $\mathbf{W}_2$ are the same magnitude w because they represent the same rigid link in two different positions. The same can be said about vectors $\mathbf{Z}_1$ and $\mathbf{Z}_2$ whose common magnitude is z .

Equations 5.5 are vector equations, each of which contains two scalar equations and so can be solved for two unknowns. The two scalar equations can be revealed by substituting Euler's identity (equation 4.4a) and separating the real and imaginary terms as was done in Section 4.5.

real part:

$$\begin{aligned} [w \cos \theta](\cos \beta_2 - 1) - [w \sin \theta] \sin \beta_2 \\ + [z \cos \phi](\cos \alpha_2 - 1) - [z \sin \phi] \sin \alpha_2 = p_{21} \cos \delta_2 \end{aligned} \quad (5.6a)$$

imaginary part (with complex operator j divided out):

$$\begin{aligned} [w \sin \theta](\cos \beta_2 - 1) + [w \cos \theta] \sin \beta_2 \\ + [z \sin \phi](\cos \alpha_2 - 1) + [z \cos \phi] \sin \alpha_2 = p_{21} \sin \delta_2 \end{aligned} \quad (5.6b)$$

There are eight variables in these two equations: w , θ , β_2 , z , ϕ , α_2 , p_{21} , and δ_2 . We can only solve for two. Three of the eight are defined in the problem statement, namely α_2 , p_{21} , and δ_2 . Of the remaining five, w , θ , β_2 , z , ϕ , we are forced to choose three as “free choices” (assumed values) in order to solve for the other two.

One strategy is to assume values for the three angles, θ , β_2 , ϕ , on the premise that we may want to specify the orientation θ , ϕ of the two link vectors $\mathbf{W}_1$ and $\mathbf{Z}_1$ to suit packaging constraints, and also specify the angular excursion β_2 of link 2 to suit some driving constraint. This choice also has the advantage of leading to a set of equations that are linear in the unknowns and are thus easy to solve. For this solution, the equations can be simplified by setting the assumed and specified terms to be equal to some constants.

In equation 5.6a, let:

$$\begin{aligned} A &= \cos \theta (\cos \beta_2 - 1) - \sin \theta \sin \beta_2 \\ B &= \cos \phi (\cos \alpha_2 - 1) - \sin \phi \sin \alpha_2 \\ C &= p_{21} \cos \delta_2 \end{aligned} \quad (5.7a)$$

and in equation 5.6b let:

$$\begin{aligned} D &= \sin \theta (\cos \beta_2 - 1) + \cos \theta \sin \beta_2 \\ E &= \sin \phi (\cos \alpha_2 - 1) + \cos \phi \sin \alpha_2 \\ F &= p_{21} \sin \delta_2 \end{aligned} \quad (5.7b)$$

then:

$$\begin{aligned} Aw + Bz &= C \\ Dw + Ez &= F \end{aligned} \quad (5.7c)$$

and solving simultaneously,

$$w = \frac{CE - BF}{AE - BD}; \quad z = \frac{AF - CD}{AE - BD} \quad (5.7d)$$

A second strategy is to assume a length z and angle ϕ for vector $\mathbf{Z}_1$ and the angular excursion β_2 of link 2 and then solve for the vector $\mathbf{W}_1$. This is a commonly used approach. Note that the terms in square brackets in each of equations 5.6 are respectively the x and y components of the vectors $\mathbf{W}_1$ and $\mathbf{Z}_1$.

$$\begin{aligned} W_{1x} &= w \cos \theta; & Z_{1x} &= z \cos \phi \\ W_{1y} &= w \sin \theta; & Z_{1y} &= z \sin \phi \end{aligned} \quad (5.8a)$$

Substituting in equation 5.6,

$$\begin{aligned} W_{1x} (\cos \beta_2 - 1) - W_{1y} \sin \beta_2 \\ + Z_{1x} (\cos \alpha_2 - 1) - Z_{1y} \sin \alpha_2 &= p_{21} \cos \delta_2 \\ W_{1y} (\cos \beta_2 - 1) + W_{1x} \sin \beta_2 \\ + Z_{1y} (\cos \alpha_2 - 1) + Z_{1x} \sin \alpha_2 &= p_{21} \sin \delta_2 \end{aligned} \quad (5.8b)$$

Z_{1x} and Z_{1y} are known from equation 5.8a with z and ϕ assumed as free choices. To further simplify the expression, combine other known terms as:

$$\begin{aligned} A &= \cos \beta_2 - 1; & B &= \sin \beta_2; & C &= \cos \alpha_2 - 1 \\ D &= \sin \alpha_2; & E &= p_{21} \cos \delta_2; & F &= p_{21} \sin \delta_2 \end{aligned} \quad (5.8c)$$

substituting,

$$\begin{aligned} AW_{1x} - BW_{1y} + CZ_{1x} - DZ_{1y} &= E \\ AW_{1y} + BW_{1x} + CZ_{1y} + DZ_{1x} &= F \end{aligned} \quad (5.8d)$$

and the solution is:

$$\begin{aligned} W_{1x} &= \frac{A(-CZ_{1x} + DZ_{1y} + E) + B(-CZ_{1y} - DZ_{1x} + F)}{-2A} \\ W_{1y} &= \frac{A(-CZ_{1y} - DZ_{1x} + F) + B(CZ_{1x} - DZ_{1y} - E)}{-2A} \end{aligned} \quad (5.8e)$$

Either of these strategies results in the definition of a left dyad $\mathbf{W}_1\mathbf{Z}_1$ and its pivot locations which will provide the motion generation specified.

We must repeat the process for the right-hand dyad, $\mathbf{U}_1\mathbf{S}_1$. Figure 5-2 highlights the two positions $\mathbf{U}_1\mathbf{S}_1$ and $\mathbf{U}_2\mathbf{S}_2$ of the right dyad. Vector $\mathbf{U}_1$ is initially at angle σ and moves through angle γ_2 from position 1 to 2. Vector $\mathbf{S}_1$ is initially at angle ψ . Note that the rotation of vector $\mathbf{S}$ from $\mathbf{S}_1$ to $\mathbf{S}_2$ is through the same angle α_2 as vector $\mathbf{Z}$, since they are in the same link. A vector loop equation similar to equation 5.3 can be written for this dyad.

$$\mathbf{U}_2 + \mathbf{S}_2 - \mathbf{P}_{21} - \mathbf{S}_1 - \mathbf{U}_1 = 0 \quad (5.9a)$$

Rewrite in complex variable form and collect terms.

$$ue^{j\sigma}(e^{j\gamma_2} - 1) + se^{j\psi}(e^{j\alpha_2} - 1) = p_{21}e^{j\delta_2} \quad (5.9b)$$

When this is expanded and the proper angles substituted, the x and y component equations become:

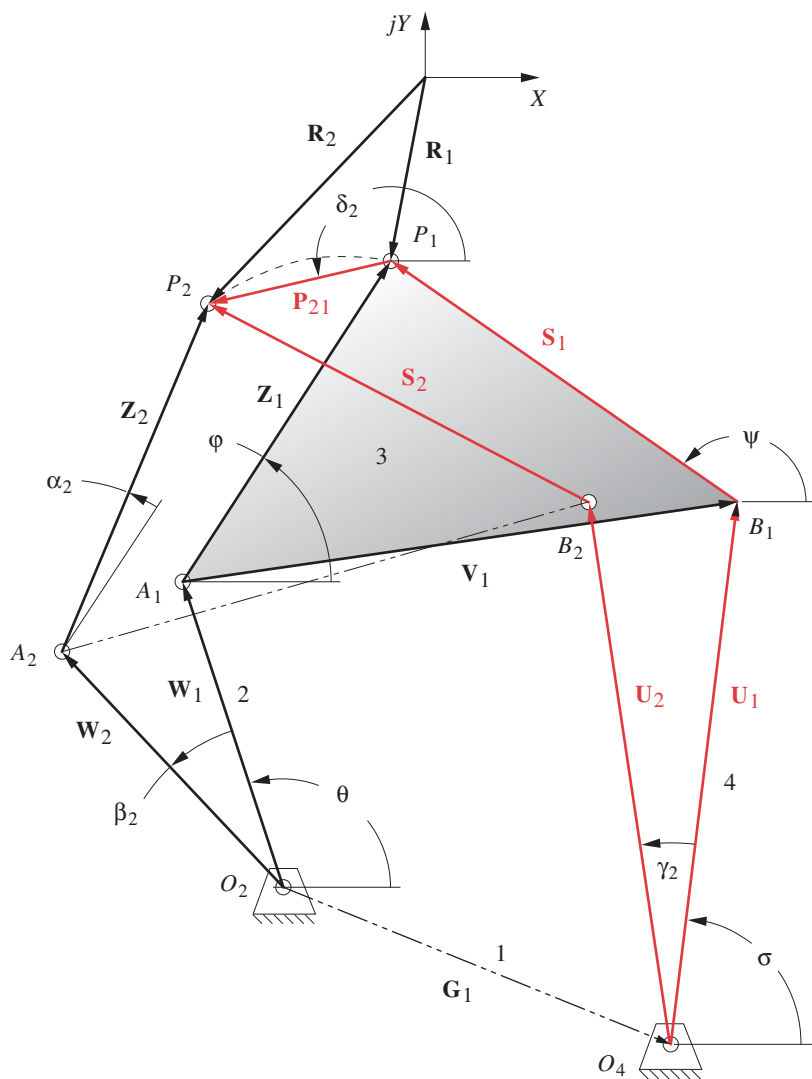
real part:

$$\begin{aligned} u \cos \sigma (\cos \gamma_2 - 1) - u \sin \sigma \sin \gamma_2 \\ + s \cos \psi (\cos \alpha_2 - 1) - s \sin \psi \sin \alpha_2 = p_{21} \cos \delta_2 \end{aligned} \quad (5.10a)$$

imaginary part (with complex operator j divided out):

$$\begin{aligned} u \sin \sigma (\cos \gamma_2 - 1) + u \cos \sigma \sin \gamma_2 \\ + s \sin \psi (\cos \alpha_2 - 1) + s \cos \psi \sin \alpha_2 = p_{21} \sin \delta_2 \end{aligned} \quad (5.10b)$$

Compare equations 5.10 to equations 5.6.

**FIGURE 5-2**

Right-side dyad shown in two positions

The same first strategy can be applied to equations 5.10 as was used for equations 5.6 to solve for the magnitudes of vectors $\mathbf{U}$ and $\mathbf{S}$, assuming values for angles σ , ψ , and γ_2 . The quantities p_{21} , δ_2 , and α_2 are defined from the problem statement as before.

In equation 5.10a let:

$$\begin{aligned}
 A &= \cos \sigma (\cos \gamma_2 - 1) - \sin \sigma \sin \gamma_2 \\
 B &= \cos \psi (\cos \alpha_2 - 1) - \sin \psi \sin \alpha_2 \\
 C &= p_{21} \cos \delta_2
 \end{aligned}
 \tag{5.11a}$$

and in equation 5.10b let:

$$\begin{aligned} D &= \sin \sigma (\cos \gamma_2 - 1) + \cos \sigma \sin \gamma_2 \\ E &= \sin \psi (\cos \alpha_2 - 1) + \cos \psi \sin \alpha_2 \\ F &= p_{21} \sin \delta_2 \end{aligned} \quad (5.11b)$$

then:

$$\begin{aligned} Au + Bs &= C \\ Du + Es &= F \end{aligned} \quad (5.11c)$$

and solving simultaneously,

$$u = \frac{CE - BF}{AE - BD}; \quad s = \frac{AF - CD}{AE - BD} \quad (5.11d)$$

If the second strategy is used, assuming angle γ_2 and the magnitude and direction of vector $\mathbf{S}_1$ (which will define link 3), the result will be:

$$\begin{aligned} U_{1x} &= u \cos \sigma; & S_{1x} &= s \cos \psi \\ U_{1y} &= u \sin \sigma; & S_{1y} &= s \sin \psi \end{aligned} \quad (5.12a)$$

Substitute in equation 5.10:

$$\begin{aligned} U_{1x} (\cos \gamma_2 - 1) - U_{1y} \sin \gamma_2 \\ + S_{1x} (\cos \alpha_2 - 1) - S_{1y} \sin \alpha_2 &= p_{21} \cos \delta_2 \\ U_{1y} (\cos \gamma_2 - 1) + U_{1x} \sin \gamma_2 \\ + S_{1y} (\cos \alpha_2 - 1) + S_{1x} \sin \alpha_2 &= p_{21} \sin \delta_2 \end{aligned} \quad (5.12b)$$

$$\begin{aligned} \text{Let: } A &= \cos \gamma_2 - 1; & B &= \sin \gamma_2; & C &= \cos \alpha_2 - 1 \\ D &= \sin \alpha_2; & E &= p_{21} \cos \delta_2; & F &= p_{21} \sin \delta_2 \end{aligned} \quad (5.12c)$$

Substitute in equation 5.12b,

$$\begin{aligned} AU_{1x} - BU_{1y} + CS_{1x} - DS_{1y} &= E \\ AU_{1y} + BU_{1x} + CS_{1y} + DS_{1x} &= F \end{aligned} \quad (5.12d)$$

and the solution is:

$$\begin{aligned} U_{1x} &= \frac{A(-CS_{1x} + DS_{1y} + E) + B(-CS_{1y} - DS_{1x} + F)}{-2A} \\ U_{1y} &= \frac{A(-CS_{1y} - DS_{1x} + F) + B(CS_{1x} - DS_{1y} - E)}{-2A} \end{aligned} \quad (5.12e)$$

Note that there are infinities of possible solutions to this problem because we may choose any set of values for the three free choices of variables in this two-position case. Technically there is an infinity of solutions for each free choice. Three choices then give infinity cubed solutions! But since infinity is defined as a number larger than the largest

number you can think of, infinity cubed is not any more impressively large than just plain infinity. While not strictly correct mathematically, we will, for simplicity, refer to all of these cases as having “an infinity of solutions,” regardless of the power to which infinity may be raised as a result of the derivation. There are plenty of solutions to pick from, at any rate. *Unfortunately, not all will work.* Some will have circuit, branch, or order (CBO) defects such as toggle positions between the precision points. Others will have poor transmission angles or poor pivot locations or overlarge links. Design judgment is still most important in selecting the assumed values for your free choices. Despite their name, you must pay for those “free choices” later. Make a model!

5.5 COMPARISON OF ANALYTICAL AND GRAPHICAL TWO-POSITION SYNTHESIS

Note that in the **graphical solution** to this two-position synthesis problem (in Example 3-3 and Figure 3-6), we also had to make *three free choices* to solve the problem. The identical two-position synthesis problem from Figure 3-6 is reproduced in Figure 5-3. The approach taken in Example 3-3 used the two points *A* and *B* (as labeled in Figure 5-3)* as the attachments for the moving pivots. Figure 5-3a shows the graphical construction used to find the fixed pivots O_2 and O_4 . For the analytical solution we will use those points *A* and *B* as the joints of the two dyads **WZ** and **US**. These dyads meet at point *P*, which is the precision point. The relative position vector $\mathbf{P}_{21}$ defines the displacement of the precision point.

Note that in the graphical solution, we implicitly defined the left dyad vector **Z** by locating attachment points *A* and *B* on link 3 as shown in Figure 5-3a. This defined the two variables, z and ϕ . We also implicitly chose the value of w by selecting an arbitrary location for pivot O_2 on the perpendicular bisector. When that third choice was made, the remaining two unknowns, angles β_2 and θ , were solved for graphically at the same time, because the geometric construction was in fact a graphical “computation” for the solution of the simultaneous equations 5.8a.

The graphical and analytical methods represent two alternate solutions to the same problem. All of these problems can be solved both analytically and graphically. One method can provide a good check for the other. We will now solve this problem analytically and correlate the results with the graphical solution from Chapter 3.

* In Figure 3-6, these same points were labeled *C* and *D*.

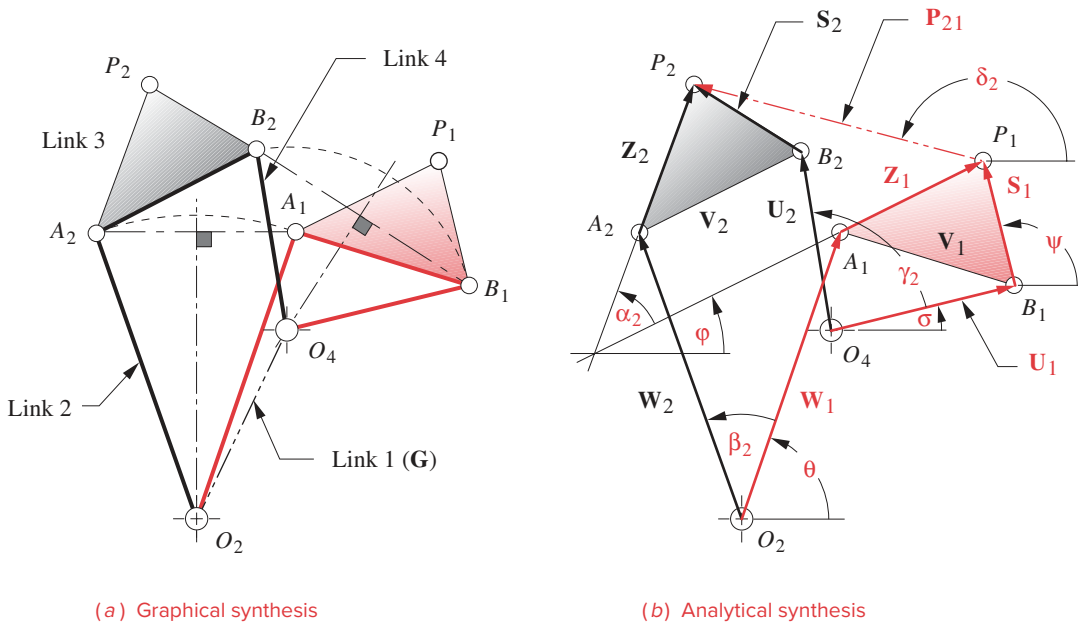
EXAMPLE 5-1

Two-Position Analytical Motion Synthesis.

Problem: Design a fourbar linkage to move the link *APB* shown from position $A_1P_1B_1$ to $A_2P_2B_2$.

Solution: (See Figure 5-3.)

- 1 Draw the link *APB* in its two desired positions, $A_1P_1B_1$ and $A_2P_2B_2$, to scale in the plane as shown.

**FIGURE 5-3**

Two-position motion synthesis with coupler output

- Measure or calculate the values of the magnitude and angle of vector $\mathbf{P}_{21}$, namely, p_{21} and δ_2 . In this example they are:

$$p_{21} = 2.416; \quad \delta_2 = 165.2^\circ$$

- Measure or calculate the value of the change in angle, α_2 , of vector $\mathbf{Z}$ from position 1 to position 2. In this example it is:

$$\alpha_2 = 43.3^\circ$$

- The three values in steps 2 and 3 are the only ones defined in the problem statement. We must assume three additional “free choices” to solve the problem. Method two (see equations 5.8) chooses the length z and angle ϕ of vector $\mathbf{Z}$ and β_2 , the change in angle of vector $\mathbf{W}$. In order to obtain the same solution as the graphical method produced in Figure 5-3a (from the infinities of solutions available), we will choose those values consistent with the graphical solution.

$$z = 1.298; \quad \phi = 26.5^\circ; \quad \beta_2 = 38.4^\circ$$

- Substitute these six values in equations 5.8 and obtain:

$$w = 2.467 \quad \theta = 71.6^\circ$$

- Compare these to the graphical solution;

$$w = 2.48 \quad \theta = 71^\circ$$

which is a reasonable match given graphical accuracy. This vector $\mathbf{W}_1$ is link 2 of the fourbar.

- 7 Repeat the procedure for the link-4 side of the linkage. The free choices will now be:

$$s = 1.035; \quad \psi = 104.1^\circ; \quad \gamma_2 = 85.6^\circ$$

- 8 Substitute these three values along with the original three values from steps 2 and 3 in equations 5.12 and obtain:

$$u = 1.486 \quad \sigma = 15.4^\circ$$

- 9 Compare these to the graphical solution:

$$u = 1.53 \quad \sigma = 14^\circ$$

These are a reasonable match for graphical accuracy. Vector $\mathbf{U}_1$ is link 4 of the fourbar.

- 10 Line A_1B_1 is link 3 and can be found from equation 5.2a. Line O_2O_4 is link 1 and can be found from equation 5.2b.
- 11 Check the Grashof condition, and repeat steps 4 to 7 if unsatisfied. Note that any Grashof condition is potentially acceptable in this case.
- 12 Construct a model in CAD or cardboard and check its function to be sure it can get from initial to final position without encountering any limit (toggle) positions.
- 13 Check transmission angles.

Open the file E05-01.4br in program LINKAGES to see Example 5-1.

5.6 SIMULTANEOUS EQUATION SOLUTION

These methods of analytical synthesis lead to sets of linear simultaneous equations. The two-position synthesis problem results in two simultaneous equations which can be solved by direct substitution. The three-position synthesis problem will lead to a system of four simultaneous linear equations and will require a more complicated method of solution. A convenient approach to the solution of sets of linear simultaneous equations is to put them in a standard matrix form and use a numerical matrix solver to obtain the answers. Matrix solvers are built into most engineering and scientific pocket calculators. Some spreadsheet packages and equation solvers will also do a matrix solution.

As an example, consider the following set of simultaneous equations:

$$\begin{array}{rrcr} -2x_1 & -x_2 & +x_3 & = & -1 \\ x_1 & +x_2 & +x_3 & = & 6 \\ 3x_1 & +x_2 & -x_3 & = & 2 \end{array} \quad (5.13a)$$

A system this small can be solved longhand by the elimination method, but we will put it in matrix form to show the general approach which will work regardless of the number of equations. The equations 5.13a can be written as the product of two matrices set equal to a third matrix.

$$\begin{bmatrix} -2 & -1 & 1 \\ 1 & 1 & 1 \\ 3 & 1 & -1 \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 6 \\ 2 \end{bmatrix} \quad (5.13b)$$

We will refer to these matrices as **A**, **B**, and **C**,

$$[\mathbf{A}] \times [\mathbf{B}] = [\mathbf{C}] \quad (5.13c)$$

where **A** is the matrix of coefficients of the unknowns, **B** is a column vector of the unknown terms, and **C** is a column vector of the constant terms. When matrix **A** is multiplied by **B**, the result will be the same as the left sides of equation 5.13a. See any text on linear algebra such as reference [7] for a discussion of the procedure for matrix multiplication.

If equation 5.13c were a scalar equation,

$$ab = c \quad (5.14a)$$

rather than a vector (matrix) equation, it would be very easy to solve it for the unknown b when a and c are known. We would simply divide c by a to find b .

$$b = \frac{c}{a} \quad (5.14b)$$

Unfortunately, division is not defined for matrices, so another approach must be used. Note that we could also express the division in equation 5.14b as:

$$b = a^{-1}c \quad (5.14c)$$

If the equations to be solved are linearly independent, then we can find the inverse of matrix **A** and multiply it by matrix **C** to find **B**. The inverse of a matrix is defined as that matrix which when multiplied by the original matrix yields the identity matrix. The **identity matrix** is a square matrix with ones on the main diagonal and zeros everywhere else. The inverse of a matrix is denoted by adding a superscript of negative one to the symbol for the original matrix.

$$[\mathbf{A}]^{-1} \times [\mathbf{A}] = [\mathbf{I}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (5.15)$$

Not all matrices will possess an inverse. The determinant of the matrix must be nonzero for an inverse to exist. The class of problems dealt with here will yield matrices which have inverses provided that all data are correctly calculated for input to the matrix and represent a real physical system. The calculation of the terms of the inverse for a matrix is a complicated numerical process which requires a computer or preprogrammed pocket calculator to invert any matrix of significant size. A Gauss-Jordan-elimination numerical method is usually used to find an inverse. For our simple example in equation 5.13 the inverse of matrix **A** is found to be:

$$\begin{bmatrix} -2 & -1 & 1 \\ 1 & 1 & 1 \\ 3 & 1 & -1 \end{bmatrix}^{-1} = \begin{bmatrix} 1.0 & 0.0 & 1.0 \\ -2.0 & 0.5 & -1.5 \\ 1. & & \end{bmatrix} \quad (5.16)$$

If the inverse of matrix $\mathbf{A}$ can be found, we can solve equations 5.13 for the unknowns $\mathbf{B}$ by multiplying both sides of the equation by the inverse of $\mathbf{A}$. Note that unlike scalar multiplication, matrix multiplication is not commutative; i.e., $\mathbf{A} \times \mathbf{B}$ is not equal to $\mathbf{B} \times \mathbf{A}$. We will premultiply each side of the equation by the inverse.

$$\begin{aligned} \text{but:} \quad & [\mathbf{A}]^{-1} \times [\mathbf{A}] \times [\mathbf{B}] = [\mathbf{A}]^{-1} \times [\mathbf{C}] \\ \text{so:} \quad & [\mathbf{A}]^{-1} \times [\mathbf{A}] = [\mathbf{I}] \\ & [\mathbf{B}] = [\mathbf{A}]^{-1} \times [\mathbf{C}] \end{aligned} \quad (5.17)$$

The product of $\mathbf{A}$ and its inverse on the left side of the equation is equal to the identity matrix $\mathbf{I}$. Multiplying by the identity matrix is equivalent, in scalar terms, to multiplying by one, so it has no effect on the result. Thus the unknowns can be found by premultiplying the inverse of the coefficient matrix $\mathbf{A}$ times the matrix of constant terms $\mathbf{C}$.

This method of solution works no matter how many equations are present as long as the inverse of $\mathbf{A}$ can be found and enough computer memory and/or time is available to do the computation. Note that it is not actually necessary to find the inverse of matrix $\mathbf{A}$ to solve the set of equations. The Gauss-Jordan algorithm which finds the inverse can also be used to directly solve for the unknowns $\mathbf{B}$ by assembling the $\mathbf{A}$ and $\mathbf{C}$ matrices into an **augmented matrix** of n rows and $n + 1$ columns. The added column is the $\mathbf{C}$ vector. This approach requires fewer calculations, so it is faster and more accurate. The augmented matrix for this example is:

$$\left[\begin{array}{ccc|c} -2 & -1 & 1 & -1 \\ 1 & 1 & 1 & 6 \\ 3 & 1 & -1 & 2 \end{array} \right] \quad (5.18a)$$

The Gauss-Jordan algorithm manipulates this augmented matrix until it is in the form shown below, in which the left, square portion has been reduced to the identity matrix and the rightmost column contains the values of the column vector of unknowns. In this case the results are $x_1 = 1$, $x_2 = 2$, and $x_3 = 3$ which are the correct solution to the original equations 5.13.

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right] \quad (5.18b)$$

The program MATRIX, supplied with this text, solves these problems with this Gauss-Jordan elimination method and operates on the augmented matrix without actually finding the inverse of $\mathbf{A}$ in explicit form. See Appendix A for instructions on running program MATRIX. For a review of matrix algebra see reference [7].

5.7 THREE-POSITION MOTION GENERATION BY ANALYTICAL SYNTHESIS

The same approach of defining two dyads, one at each end of the fourbar linkage, as used for two-position motion synthesis can be extended to three, four, and five positions in the

plane. The three-position motion synthesis problem will now be addressed. Figure 5-4 shows a fourbar linkage in one general position with a coupler point located at its first precision position P_1 . Second and third precision positions (points P_2 and P_3) are also shown. These are to be achieved by the rotation of the input rocker, link 2, through as yet unspecified angles β_2 and β_3 . Note also that the angles of the coupler link 3 at each of the precision positions are defined by the angles of the position vectors $\mathbf{Z}_1$, $\mathbf{Z}_2$, and $\mathbf{Z}_3$. The linkage shown in the figure is schematic. Its dimensions are unknown at the outset and are to be found by this synthesis technique. Thus, for example, the length of the position vector $\mathbf{Z}_1$ as shown is not indicative of the final length of that edge of link 3 nor are the lengths or angles of any of the links shown predictive of the final result.

The problem statement is:

Design a fourbar linkage which will move a line on its coupler link such that a point P on that line will be first at P_1 , later at P_2 , and still later at P_3 , and also will rotate the line through an angle α_2 between the first two precision positions and through an angle α_3 between the first and third precision positions. Find the lengths and angles of the four links and the coupler link dimensions A_1P_1 and B_1P_1 as shown in Figure 5-4.

The three-position analytical motion synthesis procedure is as follows:

For convenience, we will place the global coordinate system XY at the first precision point P_1 . We define the other two desired precision positions in the plane with respect to this global system as shown in Figure 5-4. The position difference vectors $\mathbf{P}_{21}$, drawn from P_1 to P_2 , and $\mathbf{P}_{31}$, drawn from P_1 to P_3 , have angles δ_2 and δ_3 , respectively. The position difference vectors $\mathbf{P}_{21}$ and $\mathbf{P}_{31}$ define the displacements of the output motion of point P from point 1 to 2 and from 1 to 3, respectively.

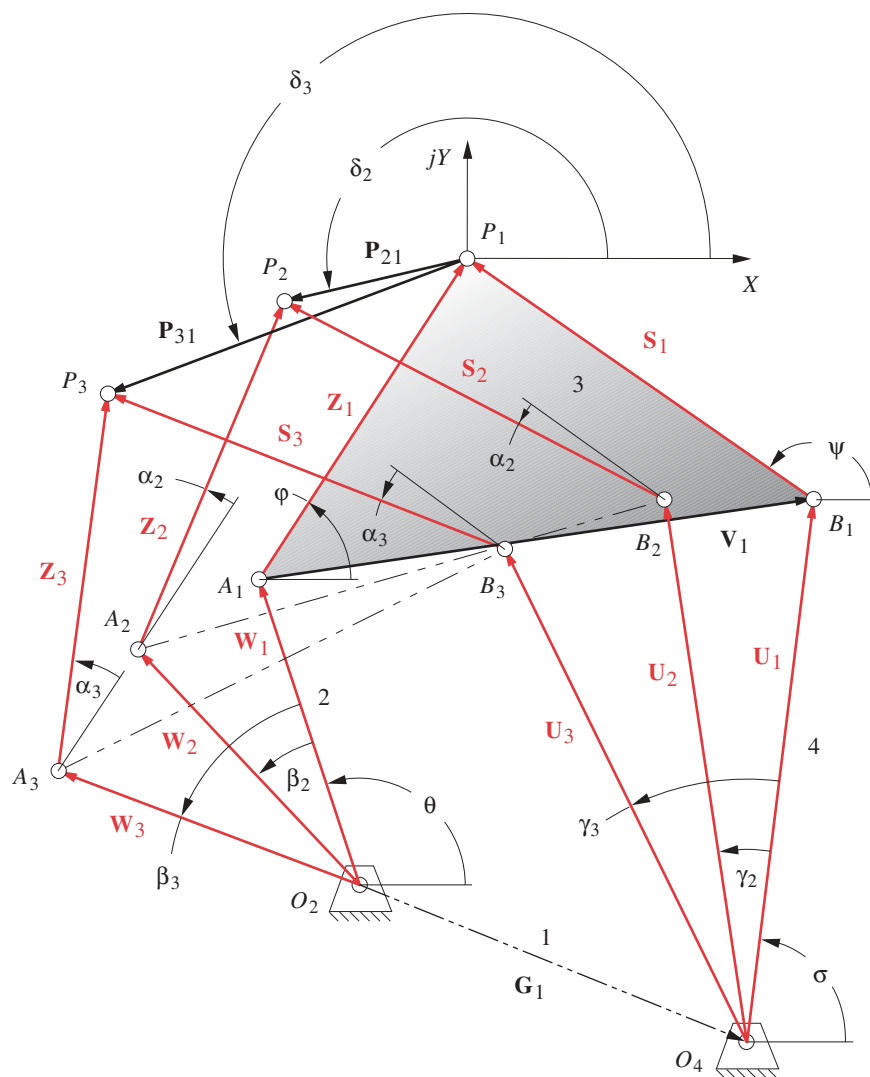
The dyad $\mathbf{W}_1\mathbf{Z}_1$ defines the left half of the linkage. The dyad $\mathbf{U}_1\mathbf{S}_1$ defines the right half of the linkage. Vectors $\mathbf{Z}_1$ and $\mathbf{S}_1$ are both embedded in the rigid coupler (link 3), and both will undergo the same rotations, through angle α_2 from position 1 to position 2 and through angle α_3 from position 1 to position 3. The pin-to-pin length and angle of link 3 (vector $\mathbf{V}_1$) are defined in terms of vectors $\mathbf{Z}_1$ and $\mathbf{S}_1$ as in equation 5.2a. The ground link is defined by equation 5.2b as before.

As we did in the two-position case, we will first solve for the left side of the linkage (vectors $\mathbf{W}_1$ and $\mathbf{Z}_1$) and later use the same procedure to solve for the right side (vectors $\mathbf{U}_1$ and $\mathbf{S}_1$). To solve for $\mathbf{W}_1$ and $\mathbf{Z}_1$ we need to now write **two vector loop equations**, one around the loop which includes positions P_1 and P_2 and the second one around the loop which includes positions P_1 and P_3 (see Figure 5-4). We will go clockwise around the first loop for motion from position 1 to 2, starting with $\mathbf{W}_2$, and then write the second loop equation for motion from position 1 to 3 starting with $\mathbf{W}_3$.

$$\begin{aligned}\mathbf{W}_2 + \mathbf{Z}_2 - \mathbf{P}_{21} - \mathbf{Z}_1 - \mathbf{W}_1 &= 0 \\ \mathbf{W}_3 + \mathbf{Z}_3 - \mathbf{P}_{31} - \mathbf{Z}_1 - \mathbf{W}_1 &= 0\end{aligned}\tag{5.19}$$

Substituting the complex number equivalents for the vectors.

$$\begin{aligned}we^{j(\theta+\beta_2)} + ze^{j(\phi+\alpha_2)} - p_{21}e^{j\delta_2} - ze^{j\phi} - we^{j\theta} &= 0 \\ we^{j(\theta+\beta_3)} + ze^{j(\phi+\alpha_3)} - p_{31}e^{j\delta_3} - ze^{j\phi} - we^{j\theta} &= 0\end{aligned}\tag{5.20}$$

**FIGURE 5-4**

Three-position analytical synthesis

Rewriting the sums of angles in the exponents as products of terms.

$$we^{j\theta}e^{j\beta_2} + ze^{j\phi}e^{j\alpha_2} - p_{21}e^{j\delta_2} - ze^{j\phi} - we^{j\theta} = 0 \quad (5.21a)$$

$$we^{j\theta}e^{j\beta_3} + ze^{j\phi}e^{j\alpha_3} - p_{31}e^{j\delta_3} - ze^{j\phi} - we^{j\theta} = 0$$

Simplifying and rearranging:

$$we^{j\theta}(e^{j\beta_2} - 1) + ze^{j\phi}(e^{j\alpha_2} - 1) = p_{21}e^{j\delta_2} \quad (5.21b)$$

$$we^{j\theta}(e^{j\beta_3} - 1) + ze^{j\phi}(e^{j\alpha_3} - 1) = p_{31}e^{j\delta_3}$$

The magnitude w of vectors $\mathbf{W}_1$, $\mathbf{W}_2$, and $\mathbf{W}_3$ is the same in all three positions because it represents the same line in a rigid link. The same can be said about vectors $\mathbf{Z}_1$, $\mathbf{Z}_2$, and $\mathbf{Z}_3$ whose common magnitude is z .

Equations 5.21 are a set of two vector equations, each of which contains two scalar equations. This set of four equations can be solved for four unknowns. The scalar equations can be revealed by substituting Euler's identity (equation 4.4a) and separating the real and imaginary terms as was done in the two-position example above.

real part:

$$\begin{aligned} w \cos \theta (\cos \beta_2 - 1) - w \sin \theta \sin \beta_2 \\ + z \cos \phi (\cos \alpha_2 - 1) - z \sin \phi \sin \alpha_2 = p_{21} \cos \delta_2 \end{aligned} \quad (5.22a)$$

$$\begin{aligned} w \cos \theta (\cos \beta_3 - 1) - w \sin \theta \sin \beta_3 \\ + z \cos \phi (\cos \alpha_3 - 1) - z \sin \phi \sin \alpha_3 = p_{31} \cos \delta_3 \end{aligned} \quad (5.22b)$$

imaginary part (with complex operator j divided out):

$$\begin{aligned} w \sin \theta (\cos \beta_2 - 1) + w \cos \theta \sin \beta_2 \\ + z \sin \phi (\cos \alpha_2 - 1) + z \cos \phi \sin \alpha_2 = p_{21} \sin \delta_2 \end{aligned} \quad (5.22c)$$

$$\begin{aligned} w \sin \theta (\cos \beta_3 - 1) + w \cos \theta \sin \beta_3 \\ + z \sin \phi (\cos \alpha_3 - 1) + z \cos \phi \sin \alpha_3 = p_{31} \sin \delta_3 \end{aligned} \quad (5.22d)$$

There are **twelve variables** in these four equations 5.22: w , θ , β_2 , β_3 , z , ϕ , α_2 , α_3 , p_{21} , p_{31} , δ_2 , and δ_3 . **We can solve for only four.** Six of them are defined in the problem statement, namely α_2 , α_3 , p_{21} , p_{31} , δ_2 , and δ_3 . Of the remaining six, w , θ , β_2 , β_3 , z , ϕ , **we must choose two as free choices** (assumed values) in order to solve for the other four. One strategy is to assume values for the two angles, β_2 and β_3 , on the premise that we may want to specify the angular excursions of link 2 to suit some driving constraint. (This choice also has the benefit of leading to a set of linear equations for simultaneous solution.)

This leaves the magnitudes and angles of vectors $\mathbf{W}$ and $\mathbf{Z}$ to be found (w , θ , z , ϕ). To simplify the solution, we can substitute the following relationships to obtain the x and y components of the two unknown vectors $\mathbf{W}$ and $\mathbf{Z}$, rather than their polar coordinates.

$$\begin{aligned} W_{1x} &= w \cos \theta; & Z_{1x} &= z \cos \phi \\ W_{1y} &= w \sin \theta; & Z_{1y} &= z \sin \phi \end{aligned} \quad (5.23)$$

Substituting equations 5.23 into 5.22 we obtain:

$$\begin{aligned} W_{1x} (\cos \beta_2 - 1) - W_{1y} \sin \beta_2 \\ + Z_{1x} (\cos \alpha_2 - 1) - Z_{1y} \sin \alpha_2 = p_{21} \cos \delta_2 \end{aligned} \quad (5.24a)$$

$$\begin{aligned} W_{1x} (\cos \beta_3 - 1) - W_{1y} \sin \beta_3 \\ + Z_{1x} (\cos \alpha_3 - 1) - Z_{1y} \sin \alpha_3 = p_{31} \cos \delta_3 \end{aligned} \quad (5.24b)$$

$$\begin{aligned}
 W_{1y} (\cos \beta_2 - 1) + W_{1x} \sin \beta_2 \\
 + Z_{1y} (\cos \alpha_2 - 1) + Z_{1x} \sin \alpha_2 = p_{21} \sin \delta_2
 \end{aligned} \quad (5.24c)$$

$$\begin{aligned}
 W_{1y} (\cos \beta_3 - 1) + W_{1x} \sin \beta_3 \\
 + Z_{1y} (\cos \alpha_3 - 1) + Z_{1x} \sin \alpha_3 = p_{31} \sin \delta_3
 \end{aligned} \quad (5.24d)$$

These are four equations in the four unknowns W_{1x} , W_{1y} , Z_{1x} , and Z_{1y} . By setting the coefficients which contain the assumed and specified terms equal to some constants, we can simplify the notation and obtain the following solutions.

$$\begin{aligned}
 A &= \cos \beta_2 - 1; & B &= \sin \beta_2; & C &= \cos \alpha_2 - 1 \\
 D &= \sin \alpha_2; & E &= p_{21} \cos \delta_2; & F &= \cos \beta_3 - 1 \\
 G &= \sin \beta_3; & H &= \cos \alpha_3 - 1; & K &= \sin \alpha_3 \\
 L &= p_{31} \cos \delta_3; & M &= p_{21} \sin \delta_2; & N &= p_{31} \sin \delta_3
 \end{aligned} \quad (5.25)$$

Substituting equations 5.25 in 5.24 to simplify:

$$AW_{1x} - BW_{1y} + CZ_{1x} - DZ_{1y} = E \quad (5.26a)$$

$$FW_{1x} - GW_{1y} + HZ_{1x} - KZ_{1y} = L \quad (5.26b)$$

$$BW_{1x} + AW_{1y} + DZ_{1x} + CZ_{1y} = M \quad (5.26c)$$

$$GW_{1x} + FW_{1y} + KZ_{1x} + HZ_{1y} = N \quad (5.26d)$$

This system can be put into standard matrix form:

$$\begin{bmatrix} A & -B & C & -D \\ F & -G & H & -K \\ B & A & D & C \\ G & F & K & H \end{bmatrix} \times \begin{bmatrix} W_{1x} \\ W_{1y} \\ Z_{1x} \\ Z_{1y} \end{bmatrix} = \begin{bmatrix} E \\ L \\ M \\ N \end{bmatrix} \quad (5.27)$$

This is the general form of equation 5.13c. The vector of unknowns $\mathbf{B}$ can be solved for by premultiplying the inverse of the coefficient matrix $\mathbf{A}$ by the constant vector $\mathbf{C}$ or by forming the augmented matrix as in equation 5.18. For any numerical problem, the inverse of a 4×4 matrix can be found with many pocket calculators. The computer program MATRIX, supplied with this text, will also solve the augmented matrix equation.

Equations 5.25 and 5.26 solve the three-position synthesis problem for the left-hand side of the linkage using any pair of assumed values for β_2 and β_3 . We must repeat the above process for the right-hand side of the linkage to find vectors $\mathbf{U}$ and $\mathbf{S}$. Figure 5-4 also shows the three positions of the $\mathbf{US}$ dyad, and the angles $\sigma, \gamma_2, \gamma_3, \psi, \alpha_2$, and α_3 , which define those vector rotations for all three positions. The solution derivation for the right-side dyad, $\mathbf{US}$, is identical to that just done for the left dyad $\mathbf{WZ}$. The angles and vector labels are the only difference. The vector loop equations are:

$$\begin{aligned}
 \mathbf{U}_2 + \mathbf{S}_2 - \mathbf{P}_{21} - \mathbf{S}_1 - \mathbf{U}_1 &= 0 \\
 \mathbf{U}_3 + \mathbf{S}_3 - \mathbf{P}_{31} - \mathbf{S}_1 - \mathbf{U}_1 &= 0
 \end{aligned} \quad (5.28)$$

Substituting, simplifying, and rearranging,

$$\begin{aligned} ue^{j\sigma}(e^{j\gamma_2} - 1) + se^{j\psi}(e^{j\alpha_2} - 1) &= p_{21}e^{j\delta_2} \\ ue^{j\sigma}(e^{j\gamma_3} - 1) + se^{j\psi}(e^{j\alpha_3} - 1) &= p_{31}e^{j\delta_3} \end{aligned} \quad (5.29)$$

The solution requires that two free choices be made. We will assume values for the angles γ_2 and γ_3 . Note that α_2 and α_3 are the same as for dyad **WZ**. We will, in effect, solve for angles σ and ψ by finding the x and y components of the vectors **U** and **S**. The solution is:

$$\begin{aligned} A &= \cos \gamma_2 - 1; & B &= \sin \gamma_2; & C &= \cos \alpha_2 - 1 \\ D &= \sin \alpha_2; & E &= p_{21} \cos \delta_2; & F &= \cos \gamma_3 - 1 \\ G &= \sin \gamma_3; & H &= \cos \alpha_3 - 1; & K &= \sin \alpha_3 \\ L &= p_{31} \cos \delta_3; & M &= p_{21} \sin \delta_2; & N &= p_{31} \sin \delta_3 \end{aligned} \quad (5.30)$$

$$AU_{1x} - BU_{1y} + CS_{1x} - DS_{1y} = E \quad (5.31a)$$

$$FU_{1x} - GU_{1y} + HS_{1x} - KS_{1y} = L \quad (5.31b)$$

$$BU_{1x} + AU_{1y} + DS_{1x} + CS_{1y} = M \quad (5.31c)$$

$$GU_{1x} + FU_{1y} + KS_{1x} + HS_{1y} = N \quad (5.31d)$$

Equations 5.31 can be solved using the approach of equations 5.27 and 5.18, by changing W to U and Z to S and using the definitions of the constants given in equation 5.30 in equation 5.27.

It should be apparent that there are infinities of solutions to this three-position synthesis problem as well. An inappropriate selection of the two free choices could lead to a solution which has circuit, branch, or order problems in moving among all specified positions. Thus we must check the function of the solution synthesized by this or any other method. A simple model is the quickest check.

5.8 COMPARISON OF ANALYTICAL AND GRAPHICAL THREE-POSITION SYNTHESIS

Figure 5-5 shows the same three-position synthesis problem as was done graphically in Example 3-6. Compare this figure to Figure 3-9. The labeling has been changed to be consistent with the notation in this chapter. The points P_1 , P_2 , and P_3 correspond to the three points labeled D in the earlier figure. Points A_1 , A_2 , and A_3 correspond to points E ; points B_1 , B_2 , and B_3 correspond to points F . The line AP becomes the **Z** vector. Point P is the coupler point which will go through the specified precision points, P_1 , P_2 , and P_3 . Points A and B are the attachment points for the rockers (links 2 and 4, respectively) on the coupler (link 3). We wish to solve for the coordinates of vectors **W**, **Z**, **U**, and **S**, which define not only the lengths of those links but also the locations of the fixed pivots O_2 and O_4 in the plane and the lengths of links 3 and 1. Link 1 is defined as vector **G** in Figure 5-4 and can be found from equation 5.2b. Link 3 is vector **V** found from equation 5.2a.

Four free choices must be made to constrain the problem to a particular solution out of the infinities of solutions available. In this case the values of link angles β_2 , β_3 , γ_2 , and γ_3 have been chosen to be the same values as those which were found in the graphical solution to Example 3-6 in order to obtain the same solution as a check and comparison. Recall that in doing the graphical three-position synthesis solution to this same problem we in fact also had to make four free choices. These were the x,y coordinates of the moving pivot locations E and F in Figure 3-9 which correspond in concept to our four free choices of link angles here.

Example 3-5 also shows a graphical solution to this same problem resulting from the free choice of the x,y coordinates of points C and D on the coupler for the moving pivots (see Figure 3-8 and Example 3-5). We found some problems with toggle positions in that solution and redid it using points E and F as moving pivots in Example 3-6 and Figure 3-9. In effect the graphical three-position synthesis solution presented in Chapter 3 is directly analogous to the analytical solution presented here. For this analytical approach we choose to select the link angles β_2 , β_3 , γ_2 , and γ_3 rather than the moving pivot locations E and F in order to force the resulting equations to be linear in the unknowns. The graphical solution done in the earlier examples is actually a solution of simultaneous nonlinear equations.



EXAMPLE 5-2

Three-Position Analytical Motion Synthesis.

Problem: Design a fourbar linkage to move the link APB shown from position $A_1P_1B_1$ to $A_2P_2B_2$ and then to position $A_3P_3B_3$.

Solution: (See Figure 5-5.)

- 1 Draw the link APB in its three desired positions, $A_1P_1B_1$, $A_2P_2B_2$, and $A_3P_3B_3$ to scale in the plane as shown in the figure.
- 2 The three positions are then defined with respect to a global origin positioned at the first precision point P_1 . The given data are the magnitudes and angles of the position difference vectors between precision points:

$$p_{21} = 2.798 \quad \delta_2 = -31.19^\circ \quad p_{31} = 3.919 \quad \delta_3 = -16.34^\circ$$

- 3 The angle changes of the coupler between precision points are:

$$\alpha_2 = -45^\circ \quad \alpha_3 = 9.3^\circ$$

- 4 The free choices assumed for the link angles are:

$$\beta_2 = 342.3^\circ \quad \beta_3 = 324.8^\circ \quad \gamma_2 = 30.9^\circ \quad \gamma_3 = 80.6^\circ$$

These defined variables and the free choices are also listed on the figure.

- 5 Once the free choices of link angles are made, the terms for the matrices of equation 5.27 can be defined by solving equation 5.25 for the first dyad of the linkage and equation 5.30 for the second dyad of the linkage. For this example they evaluate to:

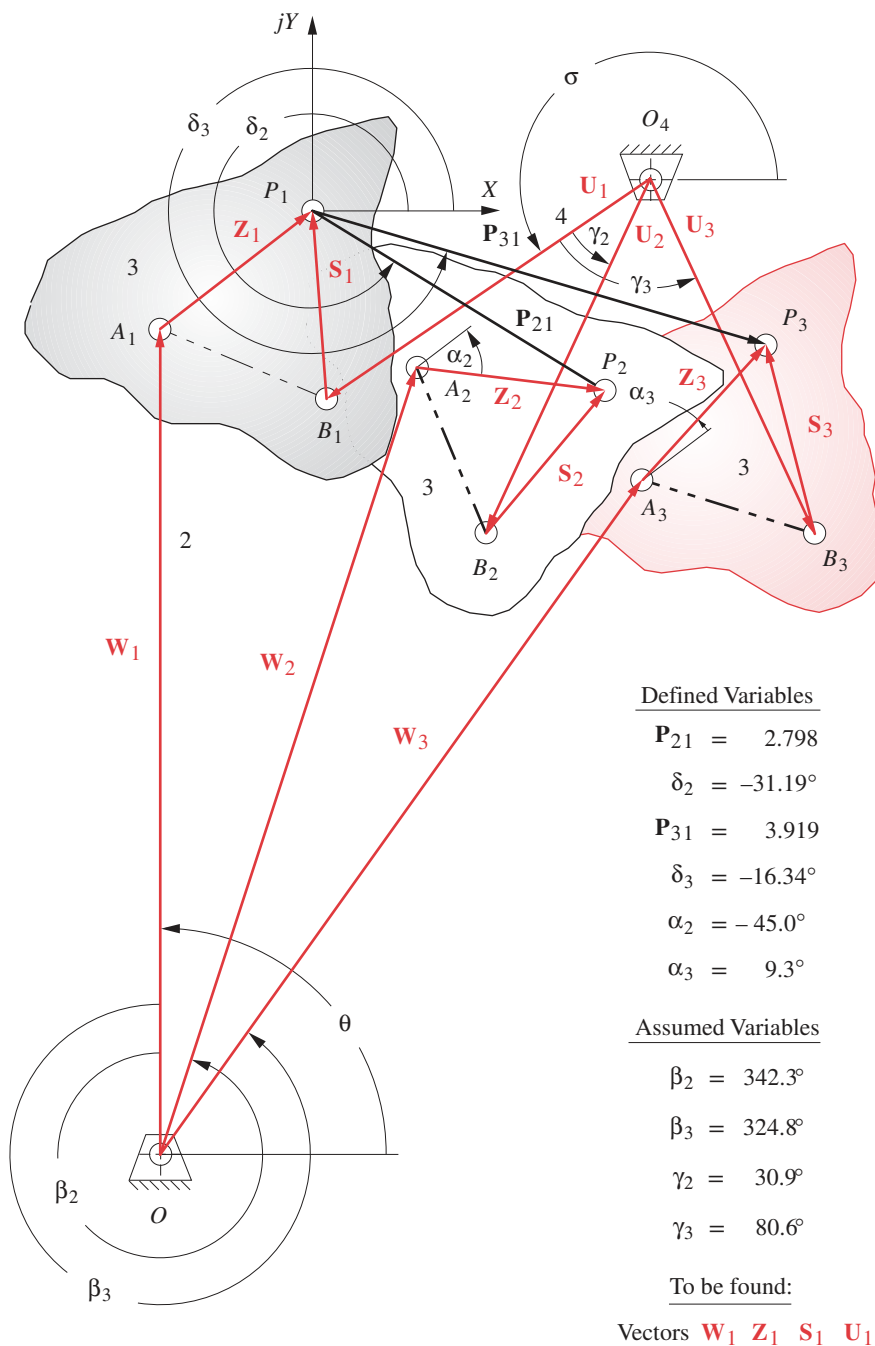


FIGURE 5-5

Data needed for three-position analytical synthesis

First dyad (**WZ**):

$$\begin{array}{llll} A = -0.0473 & B = -0.3040 & C = -0.2929 & D = -0.7071 \\ E = 2.3936 & F = -0.1829 & G = -0.5764 & H = -0.0131 \\ K = 0.1616 & L = 3.7607 & M = -1.4490 & N = -1.1026 \end{array}$$

Second dyad (**US**):

$$\begin{array}{llll} A = -0.1419 & B = 0.5135 & C = -0.2929 & D = -0.7071 \\ E = 2.3936 & F = -0.8367 & G = 0.9866 & H = -0.0131 \\ K = 0.1616 & L = 3.7607 & M = -1.4490 & N = -1.1026 \end{array}$$

- 6 Program MATRIX is used to solve this matrix equation once with the values from equation 5.25 inserted to get the coordinates of vectors **W** and **Z**, and a second time with values from equation 5.31 in the matrix to get the coordinates of vectors **U** and **S**. The calculated coordinates of the link vectors from equations 5.25 to 5.31 are:

$$W_x = 0.055 \quad W_y = 6.832 \quad Z_x = 1.179 \quad Z_y = 0.940$$

Link 2 = $w = 6.832$

$$U_x = -2.628 \quad U_y = -1.825 \quad S_x = -0.109 \quad S_y = 1.487$$

Link 4 = $u = 3.2$

- 7 Equation 5.2a is used to find link 3.

$$V_x = Z_x - S_x = 1.179 - (-0.109) = 1.288$$

$$V_y = Z_y - S_y = 0.940 - 1.487 = -0.547$$

Link 3 = $v = 1.399$

- 8 The ground link is found from equation 5.2b

$$G_x = W_x + V_x - U_x = 0.055 + 1.288 - (-2.628) = 3.971$$

$$G_y = W_y + V_y - U_y = 6.832 - 0.547 - (-1.825) = 8.110$$

Link 1 = $g = 9.03$

- 9 The appropriate vector components are added together to get the locations of fixed pivots O_2 and O_4 with respect to the global origin at precision point P_1 . See Figures 5-4 and 5-5.

$$O_{2x} = -Z_x - W_x = -1.179 - 0.055 = -1.234$$

$$O_{2y} = -Z_y - W_y = -0.940 - 6.832 = -7.772$$

$$O_{4x} = -S_x - U_x = -(-0.109) - (-2.628) = 2.737$$

$$O_{4y} = -S_y - U_y = -1.487 - (-1.825) = 0.338$$

Table 5-1 shows the linkage parameters as synthesized by this method. These agree with the solution found in Example 3-6 within its graphical accuracy. Open the files E05-02a.mtr and E05-02b.mtr in program MATRIX to compute these results.

This problem can also be solved with program LINKAGES using the same method as derived in Section 5.7. Though the derivation was done in terms of the polar coordinates of the position difference vectors $\mathbf{P}_{21}$ and $\mathbf{P}_{31}$, it was considered more convenient to supply the cartesian coordinates of these vectors to program LINKAGES. (It is generally more accurate to measure x,y coordinates from a sketch of the desired positions than to measure angles with a protractor.) Thus the program requests the rectangular coordinates of $\mathbf{P}_{21}$ and $\mathbf{P}_{31}$. For this example they are:

$p_{21x} = 2.394$ $p_{21y} = -1.449$ $p_{31x} = 3.761$ $p_{31y} = -1.103$

The angles α_2 and α_3 must be measured from the diagram and supplied, in degrees. These six items constitute the set of “givens.” **Note that these data are all relative information relating the second and third positions to the first.** No information about their absolute locations is needed. The global reference system can be taken to be anywhere in the plane. We took it to be at the first precision point position P_1 for convenience. The free choices β_2 and β_3 for the first dyad and γ_2, γ_3 for the second dyad must also be input to program LINKAGES as they also were to program MATRIX.

Program LINKAGES then solves the matrix equation 5.27 once with the values from equation 5.25 inserted to get the coordinates of vectors $\mathbf{W}$ and $\mathbf{Z}$, and a second time with values from equation 5.31 in the matrix to get the coordinates of vectors $\mathbf{U}$ and $\mathbf{S}$. Equations 5.2 are then solved to find links 1 and 3, and the appropriate vector components are added together to get the locations of fixed pivots O_2 and O_4 . The link lengths are returned to the main part of program LINKAGES so that other linkage parameters can be calculated and the linkage animated.

Note that there are two ways to assemble any fourbar linkage, open and crossed (see Figure 4-5), and this analytical synthesis technique gives no information on which mode

TABLE 5-1 Results of Analytical Synthesis for Example 5-2

Link Number	Analytical Solution Length Calculated (in)	Graphical Solution Length from Fig. 3-9 (in)
1	9.03	8.9
2	6.83	6.7
3	1.40	1.5
4	3.20	3.2
Coupler Pt. =	1.51 @ 61.31 degrees	1.5 @ 61 degrees
Open/Crossed =	CROSSED	CROSSED
Start Alpha2 =	0 rad/sec ²	
Start Omega2 =	1 rad/sec	
Start Theta2 =	29 degrees	
Final Theta2 =	11 degrees	
Delta Theta2 =	-9 degrees	

of assembly is necessary to get the desired solution. Thus you may have to try both modes of assembly in program LINKAGES to find the correct one after determining the proper link lengths with this method.

The finished linkage is the same as the one in Figure 3-9c that shows a driver dyad added to move links 2, 3, and 4 through the three precision points. You may open the file E05-02.4br in program LINKAGES to see the motions of the analytically synthesized fourbar solution. The linkage will move through the three positions defined in the problem statement. The file F03-09c.6br may also be opened in program LINKAGES to see the full motion of the finished sixbar linkage.

5.9 SYNTHESIS FOR A SPECIFIED FIXED PIVOT LOCATION

In Example 3-8 we used graphical synthesis techniques and inversion to create a fourbar linkage for three-position motion generation with specified fixed pivot locations. This is a commonly encountered problem as the available locations for fixed pivots in most machines are quite limited. Loerch et al.^[4] show how we can use these analytical synthesis techniques to find a linkage with specified fixed pivots and three output positions for motion generation. In effect we will now take as our four free choices the x and y coordinates of the two fixed pivots instead of the angles of the links. This approach will lead to a set of nonlinear equations containing transcendental functions of the unknown angles.

Figure 5-6 shows the **WZ** dyad in three positions. Because we want to relate the fixed pivots of vectors **W** and **U** to our precision points, we will place the origin of our global axis system at precision point P_1 . A position vector $\mathbf{R}_1$ can then be drawn from the root of vector $\mathbf{W}_1$ to the global origin at P_1 , $\mathbf{R}_2$ to P_2 , and $\mathbf{R}_3$ to P_3 . The vector $-\mathbf{R}_1$ defines the location of the fixed pivot in the plane with respect to the global origin at P_1 .

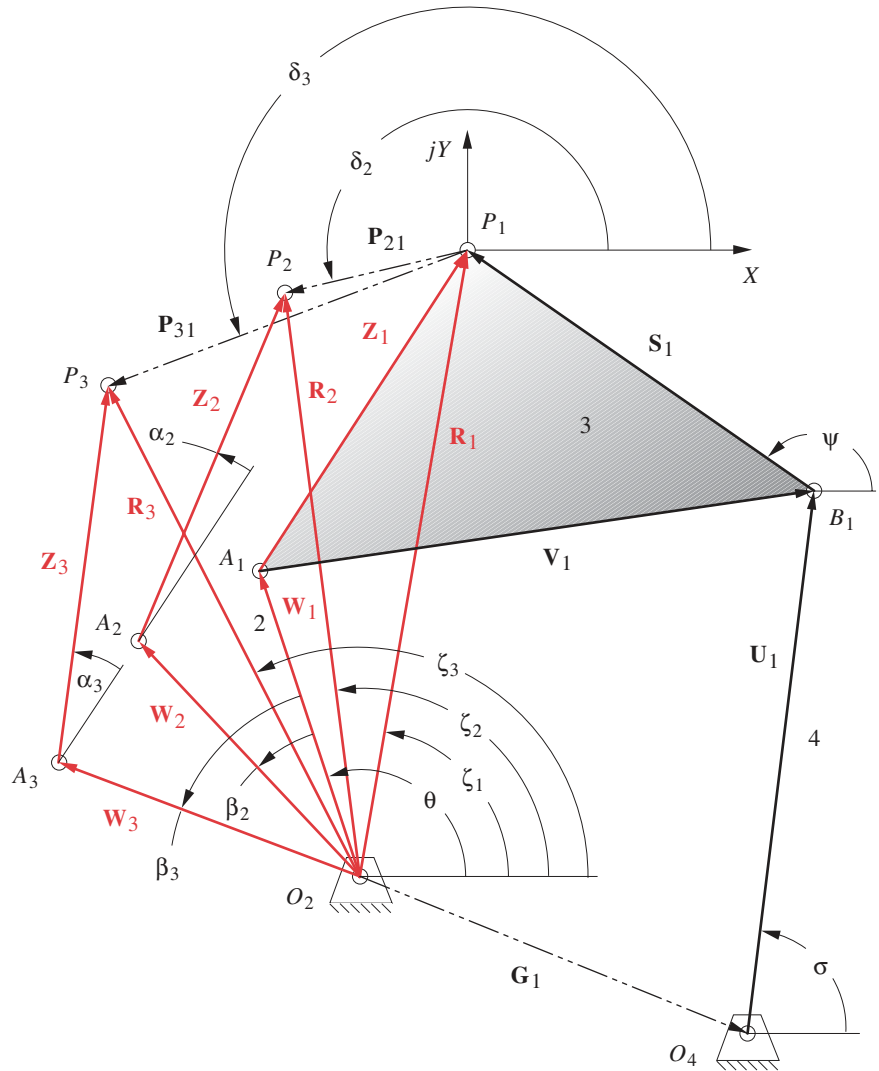
We will subsequently have to repeat this procedure for three positions of vector **U** at the right end of the linkage as we did with the three-position solution in Section 5.8. The procedure is presented here in detail only for the left end of the linkage (vectors **W**, **Z**). It is left to the reader to substitute **U** for **W** and **S** for **Z** in equations 5.32 to generate the solution for the right side.

We can write the vector loop equation for each precision position:

$$\begin{aligned}\mathbf{W}_1 + \mathbf{Z}_1 &= \mathbf{R}_1 \\ \mathbf{W}_2 + \mathbf{Z}_2 &= \mathbf{R}_2 \\ \mathbf{W}_3 + \mathbf{Z}_3 &= \mathbf{R}_3\end{aligned}\tag{5.32a}$$

Substitute the complex number equivalents for the vectors $\mathbf{W}_i$ and $\mathbf{Z}_i$:

$$\begin{aligned}we^{j\theta} + ze^{j\phi} &= \mathbf{R}_1 \\ we^{j(\theta+\beta_2)} + ze^{j(\phi+\alpha_2)} &= \mathbf{R}_2 \\ we^{j(\theta+\beta_3)} + ze^{j(\phi+\alpha_3)} &= \mathbf{R}_3\end{aligned}\tag{5.32b}$$

**FIGURE 5-6**

Three-position synthesis of a linkage with specified fixed pivot locations

Expand:

$$\begin{aligned}
 we^{j\theta} + ze^{j\phi} &= \mathbf{R}_1 \\
 we^{j\theta}e^{j\beta_2} + ze^{j\phi}e^{j\alpha_2} &= \mathbf{R}_2 \\
 we^{j\theta}e^{j\beta_3} + ze^{j\phi}e^{j\alpha_3} &= \mathbf{R}_3
 \end{aligned} \tag{5.32c}$$

Note that:

$$\mathbf{W} = we^{j\theta}; \quad \mathbf{Z} = ze^{j\phi} \tag{5.32d}$$

and:

$$\begin{aligned}\mathbf{W} + \mathbf{Z} &= \mathbf{R}_1 \\ \mathbf{W}e^{j\beta_2} + \mathbf{Z}e^{j\alpha_2} &= \mathbf{R}_2 \\ \mathbf{W}e^{j\beta_3} + \mathbf{Z}e^{j\alpha_3} &= \mathbf{R}_3\end{aligned}\quad (5.32e)$$

Previously, we chose β_2 and β_3 and solved for the vectors $\mathbf{W}$ and $\mathbf{Z}$. Now we wish to, in effect, specify the x, y components of the fixed pivot O_2 ($-R_{1x}, -R_{1y}$) as our two free choices. This leaves β_2 and β_3 to be solved for. These angles are contained in transcendental expressions in the equations. Note that, if we assumed values for β_2 and β_3 as before, there could only be a solution for $\mathbf{W}$ and $\mathbf{Z}$ if the determinant of the augmented matrix of coefficients of equations 5.32e were equal to zero.

$$\begin{vmatrix} 1 & 1 & \mathbf{R}_1 \\ e^{j\beta_2} & e^{j\alpha_2} & \mathbf{R}_2 \\ e^{j\beta_3} & e^{j\alpha_3} & \mathbf{R}_3 \end{vmatrix} = 0 \quad (5.33a)$$

Expand this determinant about the first column which contains the present unknowns β_2 and β_3 :

$$(\mathbf{R}_3e^{j\alpha_2} - \mathbf{R}_2e^{j\alpha_3}) + e^{j\beta_2}(\mathbf{R}_1e^{j\alpha_3} - \mathbf{R}_3) + e^{j\beta_3}(\mathbf{R}_2 - \mathbf{R}_1e^{j\alpha_2}) = 0 \quad (5.33b)$$

To simplify, let:

$$\begin{aligned}A &= \mathbf{R}_3e^{j\alpha_2} - \mathbf{R}_2e^{j\alpha_3} \\ B &= \mathbf{R}_1e^{j\alpha_3} - \mathbf{R}_3 \\ C &= \mathbf{R}_2 - \mathbf{R}_1e^{j\alpha_2}\end{aligned}\quad (5.33c)$$

then:

$$A + Be^{j\beta_2} + Ce^{j\beta_3} = 0 \quad (5.33d)$$

Equation 5.33d expresses the summation of vectors around a closed loop. Angles β_2 and β_3 are contained within transcendental expressions making their solution cumbersome. The procedure is similar to that used for the analysis of the fourbar linkage in Section 4.5. Substitute the complex number equivalents for all vectors in equation 5.33d. Expand using the Euler identity (equation 4.4a). Separate real and imaginary terms to get two simultaneous equations in the two unknowns β_2 and β_3 . Square these expressions and add them to eliminate one unknown. Simplify the resulting mess and substitute the tangent half angle identities to get rid of the mixture of sines and cosines. It will ultimately reduce to a quadratic equation in the tangent of half the angle sought, here β_3 . β_2 can then be found by back substituting β_3 in the original equations. The results are:

$$\begin{aligned}\beta_3 &= 2\arctan\left(\frac{K_2 \pm \sqrt{K_1^2 + K_2^2 - K_3^2}}{K_1 + K_3}\right) \\ \beta_2 &= \arctan\left[\frac{-(A_3 \sin\beta_3 + A_2 \cos\beta_3 + A_4)}{-(A_5 \sin\beta_3 + A_3 \cos\beta_3 + A_6)}\right]\end{aligned}\quad (5.34a)$$

* Note that a two-argument arctangent function must be used to obtain the proper quadrants for angles β_2 and β_3 . Also, the minus signs in numerator and denominator of the equation for β_2 look like they could be canceled, but should not be. They are needed to determine the correct quadrant of β_2 in the two-argument arctangent

where:

$$\begin{aligned} K_1 &= A_2 A_4 + A_3 A_6 \\ K_2 &= A_3 A_4 + A_5 A_6 \\ K_3 &= \frac{A_1^2 - A_2^2 - A_3^2 - A_4^2 - A_6^2}{2} \end{aligned} \quad (5.34b)$$

and:

$$\begin{aligned} A_1 &= -C_3^2 - C_4^2; & A_2 &= C_3 C_6 - C_4 C_5 \\ A_3 &= -C_4 C_6 - C_3 C_5; & A_4 &= C_2 C_3 + C_1 C_4 \\ A_5 &= C_4 C_5 - C_3 C_6; & A_6 &= C_1 C_3 - C_2 C_4 \end{aligned} \quad (5.34c)$$

$$\begin{aligned} C_1 &= R_3 \cos(\alpha_2 + \zeta_3) - R_2 \cos(\alpha_3 + \zeta_2) \\ C_2 &= R_3 \sin(\alpha_2 + \zeta_3) - R_2 \sin(\alpha_3 + \zeta_2) \\ C_3 &= R_1 \cos(\alpha_3 + \zeta_1) - R_3 \cos \zeta_3 \\ C_4 &= -R_1 \sin(\alpha_3 + \zeta_1) + R_3 \sin \zeta_3 \\ C_5 &= R_1 \cos(\alpha_2 + \zeta_1) - R_2 \cos \zeta_2 \\ C_6 &= -R_1 \sin(\alpha_2 + \zeta_1) + R_2 \sin \zeta_2 \end{aligned} \quad (5.34d)$$

The ten variables in these equations are: α_2 , α_3 , β_2 , β_3 , ζ_1 , ζ_2 , ζ_3 , R_1 , R_2 , and R_3 . The constants C_1 to C_6 are defined in terms of the eight known variables, R_1 , R_2 , R_3 , ζ_1 , ζ_2 , and ζ_3 (which are the magnitudes and angles of position vectors $\mathbf{R}_1$, $\mathbf{R}_2$, and $\mathbf{R}_3$) and the angles α_2 and α_3 that define the change in angle of the coupler. See Figure 5-6 for depictions of these variables.

Note in equation 5.34a that there are two solutions for each angle (just as there were to the position analysis of the fourbar linkage in Section 4.5 and Figure 4-5). One solution in this case will be a trivial one wherein $\beta_2 = \alpha_2$ and $\beta_3 = \alpha_3$. The nontrivial solution is the one desired.

This procedure is then repeated, solving equations 5.34 for the right-hand end of the linkage using the desired location of fixed pivot O_4 to calculate the necessary angles γ_2 and γ_3 for link 4.

We have now reduced the problem to that of three-position synthesis without specified pivots as described in Section 5.7 and Example 5-2. In effect we have found the particular values of β_2 , β_3 , γ_2 , and γ_3 which correspond to the solution that uses the desired fixed pivots. The remaining task is to solve for the values of W_x , W_y , Z_x , Z_y using equations 5.25 through 5.31.



EXAMPLE 5-3

Three-Position Analytical Synthesis with Specified Fixed Pivots.

Problem: Design a fourbar linkage to move the line AP shown from position A_1P_1 to A_2P_2 and then to position A_3P_3 using fixed pivots O_2 and O_4 in the locations specified.

Solution: (See Figure 5-7.)

- 1 Draw the link AP in its three desired positions, A_1P_1 , A_2P_2 , and A_3P_3 to scale in the plane as

shown in Figure 5-7. The three positions are defined with respect to a global origin positioned at the first precision point P_1 . The given data are specified in parts 2 to 4 below.

- 2 The position difference vectors between precision points are:

$$P_{21x} = -0.244 \quad P_{21y} = 0.013 \quad P_{31x} = -0.542 \quad P_{31y} = 0.029$$

- 3 The angle changes of the coupler between precision points are:

$$\alpha_2 = -11.34^\circ \quad \alpha_3 = -22.19^\circ$$

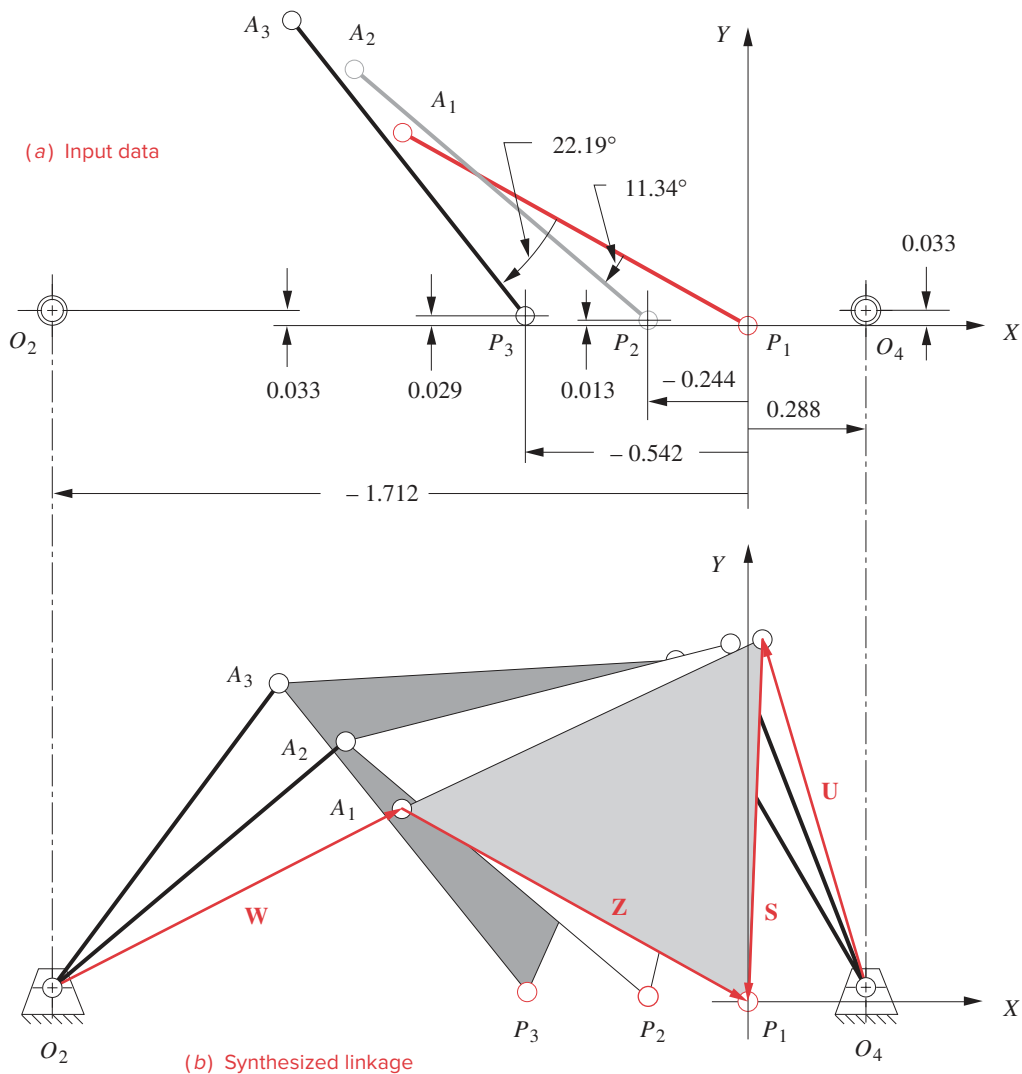


FIGURE 5-7

Three-position synthesis example for specified fixed pivots

- 4 The assumed free choices are the desired fixed pivot locations.

$$O_{2x} = -1.712 \quad O_{2y} = 0.033 \quad O_{4x} = 0.288 \quad O_{4y} = 0.033$$

- 5 Solve equations 5.34 twice, once using the O_2 pivot location coordinates and again using the O_4 pivot location coordinates.

For pivot O_2 :

$$\begin{array}{lll} C_1 = -0.205 & C_2 = 0.3390 & C_3 = 0.4028 \\ C_4 = 0.6731 & C_5 = 0.2041 & C_6 = 0.3490 \\ A_1 = -0.6152 & A_2 = 0.0032 & A_3 = -0.3171 \\ A_4 = -0.0017 & A_5 = -0.0032 & A_6 = -0.3108 \\ K_1 = 0.0986 & K_2 = 0.0015 & K_3 = 0.0907 \end{array}$$

The values found for the link angles to match this choice of fixed pivot location O_2 are:

$$\beta_2 = 11.96^\circ \quad \beta_3 = 23.96^\circ$$

For pivot O_4 :

$$\begin{array}{lll} C_1 = -0.3144 & C_2 = -0.0231 & C_3 = 0.5508 \\ C_4 = -0.0822 & C_5 = 0.2431 & C_6 = -0.0443 \\ A_1 = -0.3102 & A_2 = -0.0044 & A_3 = -0.1376 \\ A_4 = 0.0131 & A_5 = 0.0044 & A_6 = -0.1751 \\ K_1 = 0.0240 & K_2 = -0.0026 & K_3 = 0.0232 \end{array}$$

The values found for the link angles to match this choice of fixed pivot location O_4 are:

$$\gamma_2 = 2.78^\circ \quad \gamma_3 = 9.96^\circ$$

- 6 At this stage, the problem has been reduced to the same one as in the previous section; i.e., find the linkage given the free choices of the above angles β_2 , β_3 , γ_2 , γ_3 , using equations 5.25 through 5.31. The data needed for the remaining calculations are those given in steps 2, 3, and 5 of this example, namely:

for dyad 1:

$$P_{21x} \quad P_{21y} \quad P_{31x} \quad P_{31y} \quad a_2 \quad a_3 \quad b_2 \quad b_3$$

for dyad 2:

$$P_{21x} \quad P_{21y} \quad P_{31x} \quad P_{31y} \quad a_2 \quad a_3 \quad g_2 \quad g_3$$

See Example 5-2 and Section 5.7 for the procedure. A matrix solving calculator, *Mathcad*, *TKSolver*, *Matlab*, program MATRIX, or program LINKAGES will solve this and compute the coordinates of the link vectors:

$$\begin{array}{llll} W_x = 0.866 & W_y = 0.500 & Z_x = 0.846 & Z_y = -0.533 \\ U_x = -0.253 & U_y = 0.973 & S_x = -0.035 & S_y = -1.006 \end{array}$$

7 The link lengths are computed as was done in Example 5-2 and are shown in Table 5-2.

This example can be opened in program LINKAGES from the file E05-03.4br and animated.

5.10 CENTER-POINT AND CIRCLE-POINT CIRCLES

It would be quite convenient if we could find the loci of all possible solutions to the three-position synthesis problem, as we would then have an overview of the potential locations of the ends of the vectors **W**, **Z**, **U**, and **S**. Loerch et al.^[5] show that by holding one of the free choices (say β_2) at an arbitrary value and then solving equations 5.25 and 5.26 while iterating the other free choice (β_3) through all possible values from 0 to 2π , a circle will be generated. This circle is the locus of all possible locations of the root of vector **W** (for the particular value of β_2 used). The root of the vector **W** is the location of the fixed pivot or *center* O_2 . Thus, this circle is called a **center-point circle**. The vector **N** in Figure 5-8 defines points on the *center-point* circle with respect to the global coordinate system which is placed at precision point P_1 for convenience.

If the same thing is done for vector **Z**, holding α_2 constant at some arbitrary value and iterating α_3 from 0 to 2π , another circle will be generated. This circle is the locus of all possible locations of the root of vector **Z** for the chosen value of α_2 . Because the root of vector **Z** is joined to the tip of vector **W** and **W**'s tip describes a circle about pivot O_2 in the finished linkage, this locus is called the **circle-point circle**. Vector **(-Z)** defines points on the *circle-point* circle with respect to the global coordinate system.

The *x,y* components of vectors **W** and **Z** are defined by equations 5.25 and 5.26. Negating the *x,y* components of **Z** will give the coordinates of points on the circle-point circle for any assumed value of α_2 as angle α_3 is iterated from 0 to 2π . The *x,y* components of **N = -Z - W** define points on the O_2 center-point circle for any assumed value of β_2 as β_3 is iterated through 0 to 2π . Vector **W** is calculated using angles β_2 and β_3 , and vector **Z** using angles α_2 and α_3 , both from equations 5.25 and 5.26.

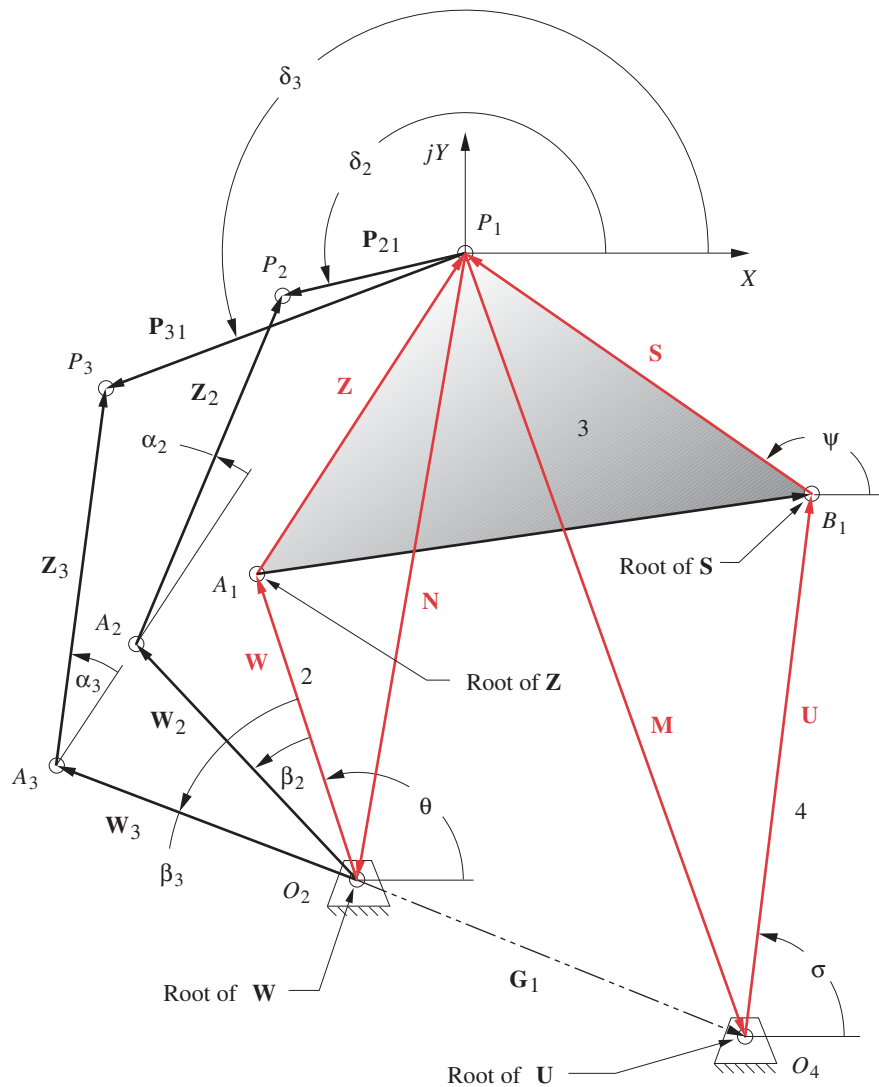
For the right-hand dyad, there will also be separate center-point circles and circle-point circles. The *x,y* components of **M = -S - U** define points on the O_4 center-point circle for any assumed value of γ_2 as γ_3 is iterated through 0 to 2π . (See Figure 5-8 and also Figure 5-4.) Negating the *x,y* components of **S** will give the coordinates of points on the circle-point circle for any assumed value of α_2 as α_3 is iterated through 0 to 2π . Vector **U** is calculated using angles γ_2 and γ_3 , and vector **S** using angles α_2 and α_3 , both from equations 5.30 and 5.31.

Note that there is still an infinity of solutions because we are choosing the value of one angle arbitrarily. Thus there will be an **infinite number of sets of center-point and circle-point circles**. A computer program can be of help in choosing a linkage design which has pivots in convenient locations. Program LINKAGES, provided with this text, will calculate the solutions to the analytical synthesis equations derived in this section, for user-selected values of all the free choices needed for three-position synthesis, both with and without specification of fixed pivot locations. Information about the computer program LINKAGES is in Appendix A.

Figure 5-9 shows the circle-point and center-point circles for the Chebyshev straight-line linkage for choices of $\beta_2 = 26^\circ$, $\alpha_2 = 97.41^\circ$, $\alpha_3 = 158.18^\circ$ for the left dyad and $\gamma =$

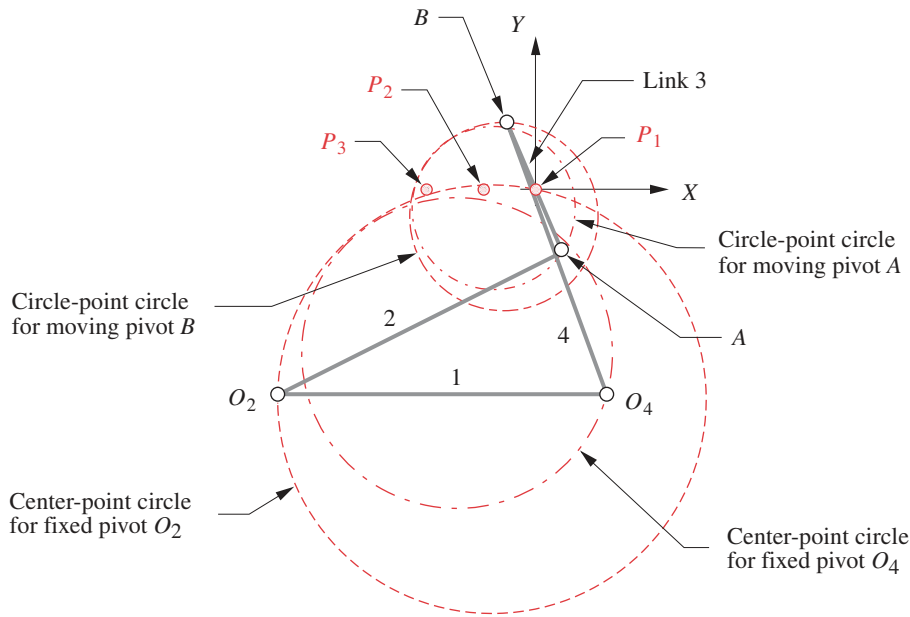
TABLE 5-2
Example 5-3 Results

Link 1 = 2.00 in
Link 2 = 1.00 in
Link 3 = 1.00 in
Link 4 = 1.01 in
Coupler Pt. =1.0 in @ -60.73°
Circuit = Open
Start Theta2 = 30°
Final Theta2 = 54°
Delta Theta2 = 12°

**FIGURE 5-8**

Definition of vectors to define center-point and circle-point circles

36° , $\alpha_2 = 97.41^\circ$, $\alpha_3 = 158.18^\circ$ for the right dyad. In this example the two larger circles are the center-point circles which define the loci of possible fixed pivot locations O_2 and O_4 . The smaller two circles define the loci of possible moving pivot locations I_{23} and I_{34} . Note that the coordinate system has its origin at the reference precision point, in this case P_1 , from which we measured all parameters used in the analysis. These circles define the pivot loci of all possible linkages which will reach the three precision points P_1 , P_2 , and P_3 that were specified for particular choices of angles β_2 , γ_2 , and α_2 . An example linkage is drawn on the diagram to illustrate one possible solution.

**FIGURE 5-9**

Circle-point and center-point circles and a linkage that reaches the precision points

5.11 FOUR- AND FIVE-POSITION ANALYTICAL SYNTHESIS

The same techniques derived above for two- and three-position synthesis can be extended to four and five positions by writing more vector loop equations, one for each precision point. To facilitate this, we will now put the vector loop equations in a more general form, applicable to any number of precision positions. Figure 5-4 will still serve to illustrate the notation for the general solution. The angles α_2 , α_3 , β_2 , β_3 , γ_2 , and γ_3 will now be designated as α_k , β_k , and γ_k , $k = 2$ to n , where k represents the precision position and $n = 2, 3, 4$, or 5 represents the total number of positions to be solved for. The vector loop general equation set then becomes:

$$\mathbf{W}_k + \mathbf{Z}_k - \mathbf{P}_{k1} - \mathbf{Z}_1 - \mathbf{W}_1 = 0, \quad k = 2 \text{ to } n \quad (5.35a)$$

which, after substituting the complex number forms and simplifying, becomes:

$$we^{j\theta} (e^{j\beta_k} - 1) + ze^{j\phi} (e^{j\alpha_k} - 1) = p_{k1}e^{j\delta_k}, \quad k = 2 \text{ to } n \quad (5.35b)$$

This can be put in a more compact form by substituting vector notation for those terms to which it applies, let:

$$\mathbf{W} = we^{j\theta}; \quad \mathbf{Z} = ze^{j\phi}; \quad \mathbf{P}_{k1} = p_{k1}e^{j\delta_k} \quad (5.35c)$$

then:

$$\mathbf{W} (e^{j\beta_k} - 1) + \mathbf{Z} (e^{j\alpha_k} - 1) = \mathbf{P}_{k1}e^{j\delta_k}, \quad k = 2 \text{ to } n \quad (5.35d)$$

Equation 5.35d is called the *standard form equation* by Erdman and Sandor.^[6] By substituting the values of α_k , β_k , and δ_k , in equation 5.35d for all the precision positions desired, the requisite set of simultaneous equations can be written for the left dyad of the linkage. The standard form equation applies to the right-hand dyad **US** as well, with appropriate changes to variable names as required.

$$\mathbf{U}(e^{j\beta_k} - 1) + \mathbf{S}(e^{j\alpha_k} - 1) = \mathbf{P}_{k1}e^{j\delta_k}, \quad k = 2 \text{ to } n \quad (5.35e)$$

The number of resulting equations, variables, and free choices for each value of n is shown in Table 5-3 (after Erdman and Sandor). They provide solutions for the four- and five-position problems in reference [6]. The circle-point and center-point circles of the three-position problem become cubic curves, called **Burmester curves**, in the four-position problem. Erdman and Gustafson's commercially available computer program LINCAGES^[8] solves the **four-position problem** in an interactive way, allowing users to select center and circle pivot locations on their Burmester curve loci, which are drawn on the graphics screen of the computer.

5.12 ANALYTICAL SYNTHESIS OF A PATH GENERATOR WITH PRESCRIBED TIMING

The approach derived above for motion generation synthesis is also applicable to the case of **path generation with prescribed timing**. In path generation, the precision points are to be reached, but the angle of a line on the coupler is not of concern. Instead, the timing at which the coupler reaches the precision point is specified in terms of input rocker angle β_2 . In the three-position motion generation problem we specified the angles α_2 and α_3 of vector **Z** in order to control the angle of the coupler. Here we instead want to specify angles β_2 and β_3 of the input rocker, to define the timing. Before, the free choices were β_2 and β_3 . Now they will be α_2 and α_3 . In either case, all four angles are either specified or assumed as free choices and the solution is identical. Figure 5-4 and equations 5.25, 5.26, 5.30, and 5.31 apply to this case as well. This case can be extended to as many as five precision points as shown in Table 5-3.

TABLE 5-3 Number of Variables and Free Choices for Analytical Precision-Point Motion and Timed Path Synthesis.^[6]

No. of Positions (n)	No. of Scalar Variables	No. of Scalar Equations	No. of Prescribed Variables	No. of Free Choices	No. of Available Solutions
2	8	2	3	3	∞^3
3	12	4	6	2	∞^2
4	16	6	9	1	∞^1
5	20	8	12	0	Finite

5.13 ANALYTICAL SYNTHESIS OF A FOURBAR FUNCTION GENERATOR

A similar process to that used for the synthesis of path generation with prescribed timing can be applied to the problem of function generation. In this case we do not care about motion of the coupler at all. In a fourbar function generator, the coupler exists only to **couple** the input link to the output link. Figure 5-10 shows a fourbar linkage in three positions. Note that the coupler, link 3, is merely a line from point A to point P . Point P can be thought of as a coupler point which happens to coincide with the pin joint between links 3 and 4. As such it will have simple arc motion, pivoting about O_4 , rather than, for example, the higher-order path motion of the coupler point P_1 in Figure 5-4.

Our **function generator** uses link 2 as the input link and takes the output from link 4. The “**function**” generated is the **relationship between the angles of link 2 and link 4** for the specified three-position positions, P_1 , P_2 , and P_3 . These are located in the plane with respect to an arbitrary global coordinate system by position vectors $\mathbf{R}_1$, $\mathbf{R}_2$, and $\mathbf{R}_3$. The function is:

$$\gamma_k = f(\beta_k), \quad k = 1, 2, \dots, n; \quad n \leq 7 \quad (5.36)$$

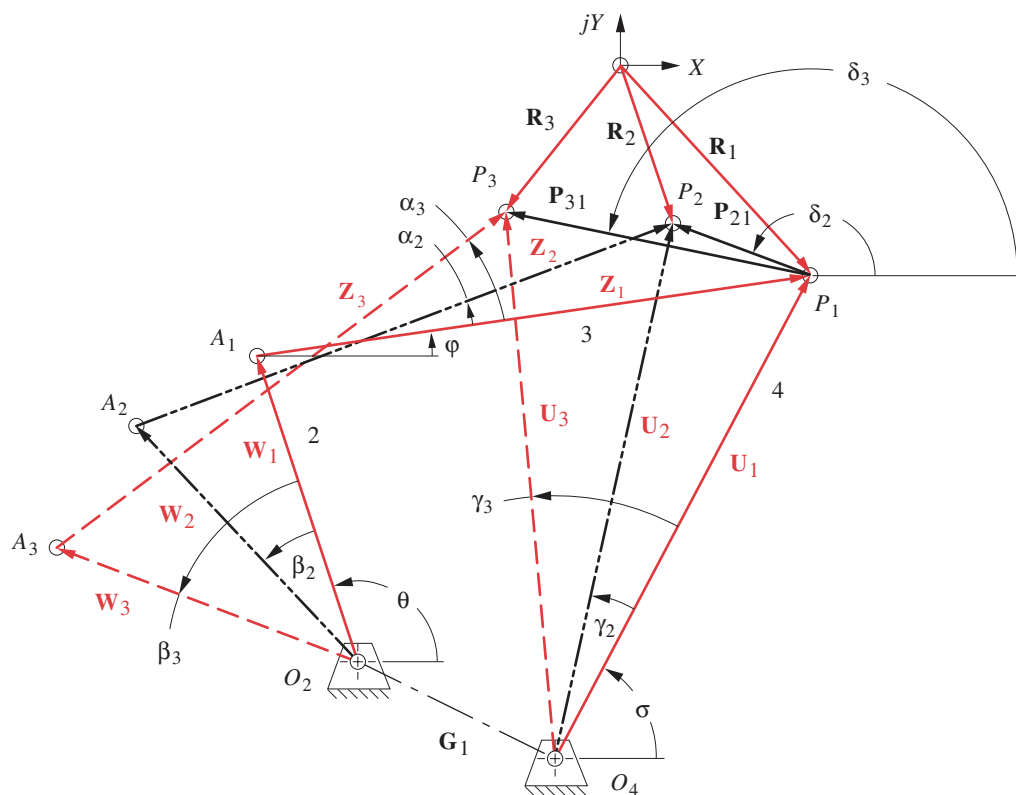


FIGURE 5-10

Analytical synthesis of a fourbar function generator

This is *not* a continuous function. The relationship holds only for the discrete points (k) specified.

To synthesize the lengths of the links needed to satisfy equation 5.36, we will write vector loop equations around the linkage in pairs of positions, as was done for the previous examples. However, we now wish to include both link 2 and link 4 in the loop, since link 4 is the output. See Figure 5-10.

$$\begin{aligned}\mathbf{W}_2 + \mathbf{Z}_2 - \mathbf{U}_2 + \mathbf{U}_1 - \mathbf{Z}_1 - \mathbf{W}_1 &= 0 \\ \mathbf{W}_3 + \mathbf{Z}_3 - \mathbf{U}_3 + \mathbf{U}_1 - \mathbf{Z}_1 - \mathbf{W}_1 &= 0\end{aligned}\quad (5.37a)$$

rearranging:

$$\begin{aligned}\mathbf{W}_2 + \mathbf{Z}_2 - \mathbf{Z}_1 - \mathbf{W}_1 &= \mathbf{U}_2 - \mathbf{U}_1 \\ \mathbf{W}_3 + \mathbf{Z}_3 - \mathbf{Z}_1 - \mathbf{W}_1 &= \mathbf{U}_3 - \mathbf{U}_1\end{aligned}\quad (5.37b)$$

but,

$$\begin{aligned}\mathbf{P}_{21} &= \mathbf{U}_2 - \mathbf{U}_1 \\ \mathbf{P}_{31} &= \mathbf{U}_3 - \mathbf{U}_1\end{aligned}\quad (5.37c)$$

substituting:

$$\begin{aligned}\mathbf{W}_2 + \mathbf{Z}_2 - \mathbf{Z}_1 - \mathbf{W}_1 &= \mathbf{P}_{21} \\ \mathbf{W}_3 + \mathbf{Z}_3 - \mathbf{Z}_1 - \mathbf{W}_1 &= \mathbf{P}_{31}\end{aligned}\quad (5.37d)$$

$$\begin{aligned}we^{j(\theta+\beta_2)} + ze^{j(\phi+\alpha_2)} - ze^{j\phi} - we^{j\theta} &= p_{21}e^{j\delta_2} \\ we^{j(\theta+\beta_3)} + ze^{j(\phi+\alpha_3)} - ze^{j\phi} - we^{j\theta} &= p_{31}e^{j\delta_3}\end{aligned}\quad (5.37e)$$

Note that equations 5.37d and 5.37e are identical to equations 5.19 and 5.20 derived for the three-position motion generation case and can also be put into Erdman's *standard form*^[6] of equation 5.35 for the n -position case. The twelve variables in equation 5.37e are the same as those in equation 5.20: w , θ , β_2 , β_3 , z , ϕ , α_2 , α_3 , p_{21} , p_{31} , δ_2 , and δ_3 .

For the three-position function generation case the solution procedure then can be the same as that described by equations 5.20 through 5.27 for the motion synthesis problem. In other words, the solution equations are the same for **all three types** of kinematic synthesis, *function generation*, *motion generation*, and *path generation with prescribed timing*. This is why Erdman and Sandor called equation 5.35 the *standard form equation*. To develop the data for the function generation solution, expand equation 5.37b:

$$\begin{aligned}we^{j(\theta+\beta_2)} + ze^{j(\phi+\alpha_2)} - ze^{j\phi} - we^{j\theta} &= ue^{j(\sigma+\gamma_2)} - ue^{j\sigma} \\ we^{j(\theta+\beta_3)} + ze^{j(\phi+\alpha_3)} - ze^{j\phi} - we^{j\theta} &= ue^{j(\sigma+\gamma_3)} - ue^{j\sigma}\end{aligned}\quad (5.37f)$$

There are also **twelve variables** in equation 5.37f: w , θ , z , ϕ , α_2 , α_3 , β_2 , β_3 , u , σ , γ_2 , and γ_3 . We can solve for any four. Four angles, β_2 , β_3 , γ_2 , and γ_3 are specified from the function to be generated in equation 5.36. This leaves **four free choices**. In the function generation problem it is often convenient to define the length of the output rocker, u , and its initial angle σ to suit the package constraints. Thus, selecting the components u and σ of vector $\mathbf{U}_1$ can provide two convenient free choices of the four required.

TABLE 5-4 Number of Variables and Free Choices for Function Generation Synthesis.^[6]

No. of Positions (n)	No. of Scalar Variables	No. of Scalar Equations	No. of Prescribed Variables	No. of Free Choices	No. of Available Solutions
2	8	2	1	5	∞^5
3	12	4	4	4	∞^4
4	16	6	7	3	∞^3
5	20	8	10	2	∞^2
6	24	10	13	1	∞^1
7	28	12	16	0	Finite

With u , σ , γ_2 , and γ_3 known, $\mathbf{U}_2$ and $\mathbf{U}_3$ can be found. Vectors $\mathbf{P}_{21}$ and $\mathbf{P}_{31}$ can then be found from equation 5.37c. Six of the unknowns in equation 5.37e are then defined, namely, β_2 , β_3 , p_{21} , p_{31} , δ_2 , and δ_3 . Of the remaining six (w , θ , z , ϕ , α_2 , α_3), we must assume values for two more as free choices in order to solve for the remaining four. We will assume values (free choices) for the two angles α_2 and α_3 (as was done for path generation with prescribed timing) and solve equations 5.37e for the components of $\mathbf{W}$ and $\mathbf{Z}$ (w , θ , z , ϕ). We have now reduced the problem to that of Section 5.7 and Example 5-2. See equations 5.20 through 5.27 for the solution.

Having chosen vector $\mathbf{U}_1$ (u , σ) as a free choice in this case, we only have to solve for one dyad, $\mathbf{WZ}$. Though we arbitrarily choose the length of vector $\mathbf{U}_1$, the resulting function generator linkage can be scaled up or down to suit packaging constraints without affecting the input/output relation defined in equation 5.36, because it is a function of angles only. This fact is not true for the motion or path generation cases, as scaling them will change the absolute coordinates of the path or motion output precision points which were specified in the problem statement.

Table 5-4 shows the relationships between number of positions, variables, free choices, and solutions for the function generation case. Note that up to seven angular output positions can be solved for with this method.

5.14 OTHER LINKAGE SYNTHESIS METHODS

Many other techniques for the synthesis of linkages to provide a prescribed motion have been created or discovered in recent years. Most of these approaches are somewhat involved and many are mathematically complicated. Only a few allow a closed-form solution; most require an iterative numerical solution. Most address the path synthesis problem with or without concern for prescribed timing. As Erdman and Sandor point out, the path, motion, and function generation problems are closely related.^[6]

Space does not permit a complete exposition of even one of these approaches in this text. We choose instead to present brief synopses of a number of synthesis methods along with complete references to their full descriptions in the engineering and scientific

literature. The reader interested in a detailed account of any method listed may consult the referenced papers which can be obtained through any university library or large public library. Also, some of the authors of these methods may make copies of their computer code available to interested parties.

Table 5-5 summarizes some of the existing fourbar linkage synthesis methods and for each one lists the method type, the maximum number of positions synthesized, the approach, special features, and a bibliographic reference (see the end of this chapter for the complete reference). The list in Table 5-5 is not exhaustive; other methods than these also exist.

The listed methods are divided into three types labeled **precision**, **equation**, and **optimized** (first column of Table 5-5). The type labeled **precision** (from precision point) refers to a method, such as the ones described in previous sections of this chapter, that attempts to find a solution that will pass exactly through the desired (precision) points but may deviate from the desired path between these points. Precision point methods are limited to matching a number of points equal to the number of independently adjustable parameters that define the mechanism. For a fourbar linkage, this is nine.* (Higher-order linkages with more links and joints will have a larger number of possible precision points.)

For up to 5 precision points in the fourbar linkage, the equations can be solved in closed form without iteration. (The four-point solution is used as a tool to solve for 5 positions in closed form, but for 6 points or more the nonlinear equations are difficult to handle.) For 6 to 9 precision points an iterative method is needed to solve the equation set. There can be problems of nonconvergence, or convergence to singular or imaginary solutions, when iterating nonlinear equations. Regardless of the number of points solved for, the solution found may be unusable due to circuit, branch, or order (CBO) defects. A circuit defect means that the linkage must be disassembled and reassembled to reach some positions, and a branch defect means that a toggle position is encountered between successive positions (see Section 4.13). An order defect means that the points are all reachable on the same branch but are encountered in the wrong order.

The type labeled **equation** in Table 5-5 refers to methods that solve the tricircular, trinodal sextic coupler curve to find a linkage that will generate an entire coupler curve that closely approximates a set of desired points on the curve.

The type labeled **optimized** in Table 5-5 refers to an iterative optimization procedure that attempts to minimize an **objective function** that can be defined in many ways, such as the least-squares deviation between the calculated and desired coupler point positions, for example. The calculated points are found by solving a set of equations that define the behavior of the linkage geometry, using assumed initial values for the linkage parameters. A set of inequality constraints that limit the range of variation of parameters such as link length ratios, Grashof condition, or transmission angle may also be included in the calculation. New values of linkage parameters are generated with each iteration step according to the particular optimization scheme used. The closest achievable fit between the calculated solution points and the desired points is sought, defined as minimization of the chosen objective function. None of the desired points will be exactly matched by these methods, but for most engineering tasks this is an acceptable result.

Optimization methods allow larger numbers of points to be specified than do the precision methods, limited only by available computer time and numerical roundoff er-

* The nine independent parameters of a fourbar linkage are: four link lengths, two coordinates of the coupler point with respect to the coupler link, and three parameters that define the location and orientation of the fixed link in the global coordinate system.

TABLE 5-5 Some Methods for the Analytic Synthesis of Linkages

Type	Max Pos.	Approach	Special Features	Bibliography	References
Precision	4	Loop equations — closed form	Linear equations extendable to five positions	Freudenstein (1959) Sandor (1959) Erdman (1981)	1, 2, 4, 5, 6, 8, 10
Precision	5	Loop equations — Newton-Raphson	Uses displacement matrix	Suh (1967)	11
Precision	5	Loop equations — continuation	Specified fixed pivots, specified moving pivots	Morgan (1990) Subbian (1991)	14, 15, 16, 17
Precision	7	Closed form 5 pt.—iterative to 7 pt.	Extendable to Watt I sixbar	Tylaska (1994)	19, 20
Precision	9	Loop equations — Newton-Raphson	Exhaustive solution	Morgan (1987) Wampler (1992)	12, 13, 18
Equation	10	Coupler curve eqn.	Iterative solution	Blechschiidt (1986)	21
Equation	15	Coupler curve eqn.	Builds on Blechschiidt	Ananthasuresh (1993)	22
Optimized	<i>N</i>	Loop equations — least squares	Specified fixed pivots, control force and torque	Fox (1966)	24
Optimized	<i>N</i>	Loop equations — various criteria	Path or function generation	Youssef (1975)	25
Optimized	<i>N</i>	Least squares on linear equations	Prescribed timing, rapid convergence	Nolle (1971)	9
Optimized	<i>N</i>	Selective precision synthesis (SPS)	Relaxes precision requirements	Kramer (1975)	26, 27
Optimized	<i>N</i>	SPS + fuzzy logic	Extends Kramer's SPS	Krishnamurthi (1993)	28
Optimized	<i>N</i>	Quasi-precision pos.	Builds on Kramer	Mirth (1994)	29
Optimized	3 or 4	Loop equations and dynamic criteria	Kinematics and dynamic forces and torques	Conte (1975) Kakatsios (1987)	30, 31, 32
Optimized	<i>N</i>	Loop equations — least squares	Avoids branch problems, rapid convergence	Angeles (1988)	33
Optimized	<i>N</i>	Energy method	FEA* approach	Aviles (1994)	34
Optimized	<i>N</i>	Genetic algorithm	Whole curve synthesis	Fang (1994)	35
Optimized	<i>N</i>	Fourier descriptors	Whole curve synthesis	Ullah (1996)	36, 37
Optimized	<i>N</i>	Neural network	Whole curve synthesis	Vasiliu (1998)	38
Optimized	2, 3, or 4	Loop equations — various criteria	Automatic generation CBO defect-free	Bawab (1997)	39
Optimized	<i>N</i>	Approximate — continuation	All solutions—no initial guess required	Liu (1999)	40

* Finite Element Analysis

ror. Table 5-5 shows a variety of optimization schemes ranging from the mundane (least squares) to the esoteric (fuzzy logic, genetic algorithms). All require a computer-programmed solution. Most can be run on current desktop computers in reasonably short times. Each different optimization approach has advantages and disadvantages in respect to convergence, accuracy, reliability, complexity, speed, and computational burden. Convergence often depends on a good choice of initial assumptions (guess values) for the

linkage parameters. Some methods, if they converge at all, do so to a local minimum (only one of many possible solutions), and it may not be the best one for the task.

Precision Point Methods

Table 5-5 shows several precision point synthesis methods. Some of these are based on original work by Freudenstein and Sandor.^[10] Sandor^[11] and Erdman^{[2], [6]} developed this approach into the “standard form” which is described in detail in this chapter. This method yields closed-form solutions for 2, 3, and 4 precision positions and is extendable to 5 positions. It suffers from the possible circuit, branch, and order (CBO) defects common to all precision point methods.

The method of Suh and Radcliffe^[11] is quite similar to that of Freudenstein and others^{[1], [2], [6], [10]} but leads to a set of simultaneous nonlinear equations which are solved for up to 5 positions using the Newton-Raphson numerical method (see Section 4.14). This approach adds to the usual CBO problems the possibilities of nonconvergence, or convergence to singular or imaginary solutions.

Recent developments in the mathematical theory of polynomials have created new methods of solution called **continuation methods** (also called **homotopy methods**) which do not suffer from the same convergence problems as other methods and can also determine all the solutions of the equations starting from any set of assumed values.^{[12], [13]} Continuation methods are a general solution to this class of problem and are reliable and fast enough to allow multiple designs to be investigated in a reasonable time (typically measured in CPU **hours** on a powerful computer).

Several researchers have developed solutions for the 5- to 9-precision point problem using this technique. Morgan and Wampler^[14] solved the fourbar linkage 5-point problem with specified fixed pivots completely and found a maximum of 36 real solutions. Subbian and Flugrad^[15] used specified moving pivots for the 5-point problem, extended the 5-point method to sixbar linkages,^[16] and also synthesized eightbar and geared fivebar mechanisms for 6 and 7 precision points using continuation methods.^[17]

Only the continuation method has yet been able to completely solve the fourbar linkage 9-precision-point problem and yield all its possible solutions. Wampler, Morgan, and Sommese^[18] used a combination of analytical equation reduction and numerical continuation methods to exhaustively compute all possible nondegenerate, generic solutions to the 9-point problem.* They proved that there is a maximum of 4326 distinct, nondegenerate linkages (occurring in 1442 sets of cognate triples) that will potentially solve a generic 9-precision-point fourbar problem. Their method does not eliminate physically impossible (complex link) linkages or those with CBO defects. These still have to be removed by examination of the various solutions. They also solved four examples and found the maximum number of linkages with real link lengths that generated these particular 9-point paths to be 21, 45, 64, and 120 cognate triples. Computation times ranged from 69 to 321 CPU minutes on an IBM 3090 for these four examples.

Tylaska and Kazerounian^{[19], [20]} took a different approach and devised a method that synthesizes a fourbar linkage for up to 7 precision points and also synthesized a Watt I sixbar linkage for up to six body guidance (motion specification) positions with control over locations of some ground and moving pivots. Their method yields the entire set of

* The authors report that this calculation took 332 CPU hours on an IBM 3081 computer.

solutions for any set of design data and is an improvement over iterative methods that are sensitive to initial guesses. It is less computationally intensive than continuation methods.

Coupler Curve Equation Methods

Bleichschmidt and Uicker^[21] and Ananthasuresh and Kota^[22] used the algebraic coupler curve equation rather than a vector loop approach to calculate the coupler point path. The equation of the coupler curve is a tricircular, trinodal sextic of 15 terms. Beyer^[41] gives one form of the coupler curve equation as:*

$$a^2 \left[(x-k)^2 + y^2 \right] (x^2 + y^2 + b^2 - r^2)^2 - 2ab \left[(x^2 + y^2 - kx) \cos \gamma + ky \sin \gamma \right] \\ \left(x^2 + y^2 + b^2 - r^2 \right) \left[(x-k)^2 + y^2 + a^2 - R^2 \right] + b^2 (x^2 + y^2) \\ \left[(x-k)^2 + y^2 + a^2 - R^2 \right]^2 - 4a^2 b^2 \left[(x^2 + y^2 - kx) \sin \gamma - ky \cos \gamma \right]^2 = 0 \quad (5.38)$$

Nolle^[23] states that:

The coupler curve equation itself is very complex and as far as is known in the study of mechanics (or for that matter elsewhere) no other mathematical result has been found having algebraic characteristics matching those of the coupler curve.

Its solution is quite involved and requires iteration. Bleichschmidt and Uicker's approach^[21] chose coordinates for 10 points on the desired curve. Ananthasuresh and Kota^[22] used 15 points with some trial and error required in their selection. The advantage of these coupler curve equation approaches is that they define the entire curve which can be plotted and examined for suitability and defects prior to calculating the link dimensions, which requires significant additional computing time.

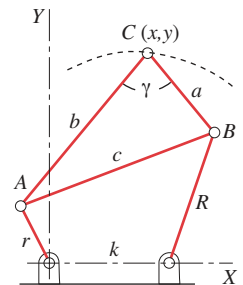
Optimization Methods

The methods listed as **optimized** in Table 5-5 are a diverse group and some have little in common except the goal of finding a linkage that will generate a desired path. All allow a theoretically unlimited number of design points to be specified, but making N too large will increase the computation time and may not improve the result. One inherent limitation of optimization methods is that they may converge to a local minimum near the starting conditions. The result may not be as good as other minima located elsewhere in the N -space of the variables. Finding the global optimum is possible but more difficult and time consuming.

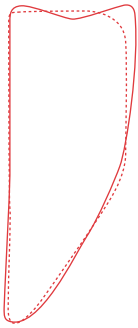
Perhaps the earliest application (1966) of optimization techniques to this fourbar linkage path synthesis problem is that of Fox and Willmert^[24] in which they minimized the area between the desired and calculated curves subject to a number of equality and inequality constraints. They controlled link lengths to be positive and less than some maximum, controlled for Grashof condition, limited forces and torques, and restricted the locations of the fixed pivots. They used Powell's method to find the minimum of the objective function.

Youssef et al.^[25] used sum of squares, sum of absolute values, or area error criteria to minimize the objective function. They accommodated path and function generation for

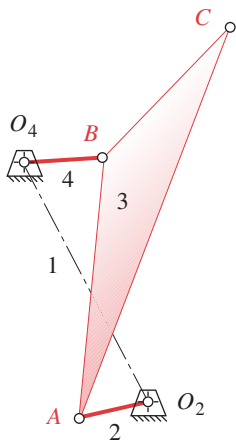
* Beyer's linkage geometry notation is different than that used in this book. Beyer's labeling for the equation, as shown by Hall,^[42] is:



desired curve
actual curve —



(a) Coupler curve



(b) Synthesized linkage

FIGURE 5-11

Linkage synthesized to generate a desired coupler curve by an optimization method
Reproduced from
"Optimal Kinematic
Synthesis of Planar Linkage
Mechanisms"^[25] with the
kind permission of
Professional Engineering
Publishing, Bury St.
Edmunds, UK.

single-loop (fourbar) or multiloop linkages with both pin and slider joints. They allowed constraints to be imposed on the allowable ranges of link lengths and angles, any of which also may be held constant during the iteration. An example of an optimization done with this method for 19 evenly spaced points around a desired fourbar coupler path is shown in Figure 5-11^[25] Another example of this method is the 10-bar crank-slider linkage in Figure 5-12^[25] which also shows the desired and actual coupler curve generated by point P for 24 points corresponding to equal increments of input crank angle.

Nolle and Hunt^[9] derived analytical expressions that lead to a set of ten linear simultaneous nonhomogeneous equations whose solution gives values for all the independent variables. They used a least squares approach to the optimization and also allowed specified timing of the input crank to each position on the coupler. Because their equations are linear, convergence is rapid requiring only about one second per iteration.

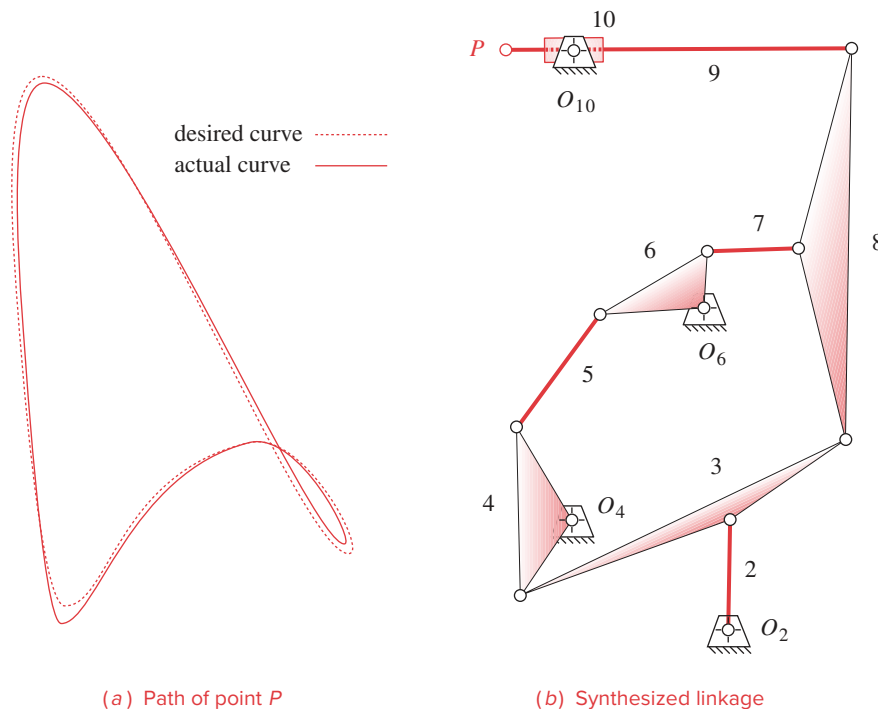
Kramer and Sandor^{[26], [27]} described a variant on the precision point technique which they call **selective precision synthesis** (SPS). It relaxes the requirement that the curve pass exactly through the precision points by defining "accuracy neighborhoods" around each point. The size of these tolerance zones can be different for each point, and more than nine points can be used. They point out that exact correspondence to a set of points is often not necessary in engineering applications and even if achieved theoretically would be compromised by manufacturing tolerances.

The SPS approach is suitable to any linkage constructible from dyads or triads and so can accommodate sixbar and geared fivebar linkages as well as fourbars. Fourbar function, motion, or path generation (with prescribed timing) can all be synthesized, using the standard form approach which considers all three forms equivalent in terms of equation formulation. Spatial mechanisms can also be accommodated. The solutions are stable and less sensitive to small changes in the data than precision point methods. Krishnamurthi et al.^[28] extended the SPS approach by using fuzzy set theory which gives a mechanism path as close to the specified points as is possible for a given start point, but it is sensitive to start point selection and may find a local, rather than the global optimum.

Mirth^[29] provided a variation on Kramer's SPS technique called quasi-precision position synthesis which uses three precision positions and N quasi positions which are defined as tolerance zones. This approach retains the computational advantages of the Burmester (precision point) approach while also allowing the specification of a larger number of points to improve and refine the design.

Conte et al.^[30] and Kakatsios and Tricamo^{[31], [32]} described methods to satisfy a small number of precision points and simultaneously optimize the linkage's dynamic characteristics. The link lengths are controlled to reasonable size, the Grashof condition is constrained, and the input torque, dynamic bearing and reaction forces, and shaking moments are simultaneously minimized.

Many of the optimization methods listed above use some form of inequality constraints to limit the allowable values of design parameters such as link lengths and transmission angles. These constraints often cause problems that lead to nonconvergence, or to CBO defects. Angeles et al.^[33] described an unconstrained nonlinear least-squares method that avoids these problems. Continuation methods are employed, and good convergence is claimed with no branch defects.

**FIGURE 5-12**

Example of synthesis of a 10 link mechanism to generate a coupler path

Reproduced from Youssef et al. (1975), "Optimal Kinematic Synthesis of Planar Linkage Mechanisms"^[25] with the kind permission of Professional Engineering Publishing, Bury St. Edmunds, UK.

Aviles et al.^[34] proposed a novel approach to the linkage synthesis problem that uses the elastic energy that would be stored in the links if they were allowed to deform elastically such that the coupler point reaches the desired location. The objective function is defined as the minimum energy condition in the set of deformed links which of course will occur when their rigid body positions most closely approach the desired path. This is essentially a finite element method approach that considers each link to be a bar element. Newton's method is used for the iteration and, in this case, converges to a minimum even when the initial guess is far from a solution.

Fang^[35] described an unusual approach to linkage synthesis using genetic algorithms. Genetic algorithms emulate the way that living organisms adapt to nature. Initially, a population of random "organisms" is generated that represents the system to be optimized. This takes the form of a bit string, analogous to a cell's chromosomes, which is called the first generation. Two operations are performed on a given population, called crossover and mutation. Crossover combines part of the "genetic code" of a "father" organism with part of the code of a "mother" organism. Mutation changes values of the genetic code at random points in the bit string. An objective function is created that

expresses the “fitness” of the organism for the desired task. Each successive generation is produced by selecting the organisms that best fit the task. The population “evolves” through generations until a termination criterion is met based on the objective function.

Some advantages of this approach are that it searches from population to population rather than point to point, and this makes it less likely to be trapped at local optima. The population also preserves a number of valid solutions rather than converging to only one. The disadvantage is long computation times due to the large number of objective function evaluations required. Nevertheless it is more efficient than random walk or exhaustive search algorithms. All other optimization approaches listed here deal only with dimensional synthesis, but genetic algorithms can also deal with type synthesis.

Ullah and Kota^{[36], [37]} separated the linkage synthesis problem into two steps. The first step seeks an acceptable match for the shape of the desired curve without regard to the size, orientation, or location of the curve in space. Once a curve of suitable shape and its associated linkage are found, the result can be translated, rotated, and scaled as desired. This approach simplifies the optimization task compared to the algorithms that seek a structural optimization that includes size, orientation, and location of the coupler curve all at once in the objective function. Fourier descriptors are used to characterize the shape of the curve as is done in many pattern matching applications such as for automated robotic assembly tasks. A stochastic global optimization algorithm is used which avoids unwanted convergence to suboptimal local minima.

Vasiliu and Yannou^[38] also focus solely on the shape of the desired path, approximating it with five terms of a Fourier series. They use an artificial *neural network* approach to synthesize a linkage to generate the approximate curve shape. A neural network is a graph of *input neurons* that represent the shape of the path and *output neurons* that represent the dimensional parameters of the linkage. The network is “taught” to properly relate the output to the input with various algorithms. Learning time was 30 hours for 14 000 iterations for their example, so this method is computer intensive. The matching of their resulting linkage curve shape to the desired curve is less accurate than that of the method shown in Figures 5-11 and 5-12.

Bawab et al.^[39] described an approach that will automatically (within the software program) synthesize a fourbar linkage for two, three, or four positions using Burmester theory and eliminate all solutions having CBO defects. Limits on link length ratios and transmission angle are specified, and the objective function is based on these criteria with weighting factors applied. Regions in the plane within which the fixed or moving pivots must be located may also be specified.

Liu and Yang^[40] proposed a method for finding all solutions to the approximate synthesis problem for function generation, rigid body guidance, and path generation with timing, using a combination of continuation methods and optimization. Their approach does not require an initial guess, and all possible solutions can be obtained with relatively short computational times.

5.15 REFERENCES

- 1 **Sandor, G. N.** (1959). "A General Complex Number Method for Plane Kinematic Synthesis with Applications." Ph.D. Thesis, Columbia University, University Microfilms, Ann Arbor, MI.
- 2 **Erdman, A. G.** (1981). "Three and Four Precision Point Kinematic Synthesis of Planar Linkages." *Mechanism and Machine Theory*, **16**, pp. 227-245.
- 3 **Kaufman, R. E.** (1978). "Mechanism Design by Computer." *Machine Design*, October 26, 1978, pp. 94-100.
- 4 **Loerch, R. J., et al.** (1975). "Synthesis of Fourbar Linkages with Specified Ground Pivots." *Proc. of 4th Applied Mechanisms Conference*, Chicago, IL, pp. 10.1-10.6.
- 5 **Loerch, R. J., et al.** (1979). "On the Existence of Circle-Point and Center-Point Circles for Three Position Dyad Synthesis." *Journal of Mechanical Design*, **101**(3), pp. 554-562.
- 6 **Erdman, A. G., and G. N. Sandor.** (1997). *Mechanism Design: Analysis and Synthesis*. Vol. 1, 3d ed., and *Advanced Mechanism Design, Analysis and Synthesis*, Vol. 2 (1984). Prentice-Hall: Upper Saddle River, NJ.
- 7 **Jennings, A.** (1977). *Matrix Computation for Engineers and Scientists*. John Wiley & Sons: New York.
- 8 **Erdman, A. G., and J. E. Gustafson.** (1977). "LINCAGES: Linkage Interactive Computer Analysis and Graphically Enhanced Synthesis." ASME Paper: 77-DTC-5.
- 9 **Nolle, H., and K. H. Hunt.** (1971). "Optimum Synthesis of Planar Linkages to Generate Coupler Curves." *Journal of Mechanisms*, **6**, pp. 267-287.
- 10 **Freudenstein, F., and G. N. Sandor.** (1959). "Synthesis of Path Generating Mechanisms by Means of a Programmed Digital Computer." *ASME Journal for Engineering in Industry*, **81**, p. 2.
- 11 **Suh, C. H., and C. W. Radcliffe.** (1966). "Synthesis of Planar Linkages With Use of the Displacement Matrix." ASME Paper: 66-MECH-19, 9 pp.
- 12 **Morgan, A. P., and A. J. Sommese.** (1987). "Computing All Solutions to Polynomial Systems Using Homotopy Continuation." *Applied Mathematics and Computation*, **24**, pp. 115-138.
- 13 **Morgan, A. P.** (1987). *Solving Polynomial Systems Using Continuation for Scientific and Engineering Problems*. Prentice-Hall: Upper Saddle River, NJ.
- 14 **Morgan, A. P., and C. W. Wampler.** (1990). "Solving a Planar Fourbar Design Problem Using Continuation." *Journal of Mechanical Design*, **112**(4), p. 544.
- 15 **Subbian, T., and J. D. R. Flugrad.** (1991). "Fourbar Path Generation Synthesis by a Continuation Method." *Journal of Mechanical Design*, **113**(1), p. 63.
- 16 **Subbian, T., and J. D. R. Flugrad.** (1993). "Five Position Triad Synthesis with Applications to Four and Sixbar Mechanisms." *Journal of Mechanical Design*, **115**(2), p. 262.
- 17 **Subbian, T., and J. D. R. Flugrad.** (1994). "Six and Seven Position Triad Synthesis Using Continuation Methods." *Journal of Mechanical Design*, **116**(2), p. 660.
- 18 **Wampler, C. W., et al.** (1992). "Complete Solution of the Nine-Point Path Synthesis Problem for Fourbar Linkages." *Journal of Mechanical Design*, **114**(1), p. 153.
- 19 **Tylaska, T., and K. Kazerounian.** (1994). "Synthesis of Defect-Free Sixbar Linkages for Body Guidance Through up to Six Finitely Separated Positions." *Proc. of 23rd Biennial Mechanisms Conference*, Minneapolis, MN, p. 369.
- 20 **Tylaska, T., and K. Kazerounian.** (1993). "Design of a Six Position Body Guidance Watt I Sixbar Linkage and Related Concepts." *Proc. of 3rd Applied Mechanisms and Robotics Conference*, Cincinnati, p. 93-1.
- 21 **Blechschildt, J. L., and J. J. Uicker.** (1986). "Linkage Synthesis Using Algebraic Curves." *J. Mechanisms, Transmissions, and Automation in Design*, **108** (December 1986), pp. 543-548.

- 22 **Ananthasuresh, G. K., and S. Kota.** (1993). "A Renewed Approach to the Synthesis of Fourbar Linkages for Path Generation via the Coupler Curve Equation." *Proc. of 3rd Applied Mechanisms and Robotics Conference*, Cincinnati, p. 83-1.
- 23 **Nolle, H.** (1975). "Linkage Coupler Curve Synthesis: A Historical Review - III. Spatial Synthesis and Optimization." *Mechanism and Machine Theory*, **10**, 1975, pp. 41-55.
- 24 **Fox, R. L., and K. D. Willmert.** (1967). "Optimum Design of Curve-Generating Linkages with Inequality Constraints." *Journal of Engineering for Industry* (Feb. 1967), pp. 144-152.
- 25 **Youssef, A. H., et al.** (1975). "Optimal Kinematic Synthesis of Planar Linkage Mechanisms." *I. Mech. E.*, pp. 393-398.
- 26 **Kramer, S. N., and G. N. Sandor.** (1975). "Selective Precision Synthesis—A General Method of Optimization for Planar Mechanisms." *Trans. ASME J. Eng. for Industry*, **97B**(2), pp. 689-701.
- 27 **Kramer, S. N.** (1987). "Selective Precision Synthesis—General Design Method for Planar and Spatial Mechanisms." *Proc. of 7th World Congress on Theory of Machines and Mechanisms*, Seville, Spain.
- 28 **Krishnamurthi, S., et al.** (1993). "Fuzzy Synthesis of Mechanisms." *Proc. of 3rd Applied Mechanisms and Robotics Conference*, Cincinnati, p. 94-1.
- 29 **Mirth, J. A.** (1994). "Quasi-Precision Position Synthesis of Fourbar Linkages." *Proc. of 23rd Biennial Mechanisms Conference*, Minneapolis, MN, p. 215.
- 30 **Conte, F. L., et al.** (1975). "Optimum Mechanism Design Combining Kinematic and Dynamic-Force Considerations." *Journal of Engineering for Industry* (May 1975), pp. 662-670.
- 31 **Kakatsios, A. J., and S. J. Tricamo.** (1987). "Precision Point Synthesis of Mechanisms with Optimal Dynamic Characteristics." *Proc. of 7th World Congress on the Theory of Machines and Mechanisms*, Seville, Spain, pp. 1041-1046.
- 32 **Kakatsios, A. J., and S. J. Tricamo.** (1986). "Design of Planar Rigid Body Guidance Mechanisms with Simultaneously Optimized Kinematic and Dynamic Characteristics." ASME Paper: 86-DET-142.
- 33 **Angeles, J., et al.** (1988). "An Unconstrained Nonlinear Least-Square Method of Optimization of RRRR Planar Path Generators." *Mechanism and Machine Theory*, **23**(5), pp. 343-353.
- 34 **Aviles, R., et al.** (1994). "An Energy-Based General Method for the Optimum Synthesis of Mechanisms." *Journal of Mechanical Design*, **116**(1), p. 127.
- 35 **Fang, W. E.** (1994). "Simultaneous Type and Dimensional Synthesis of Mechanisms by Genetic Algorithms." *Proc. of 23rd Biennial Mechanisms Conference*, Minneapolis, MN, p. 36.
- 36 **Ullah, I., and S. Kota.** (1994). "A More Effective Formulation of the Path Generation Mechanism Synthesis Problem." *Proc. of 23rd Biennial Mechanisms Conference*, Minneapolis, MN, p. 239.
- 37 **Ullah, I., and S. Kota.** (1996). "Globally-Optimal Synthesis of Mechanisms for Path Generation Using Simulated Annealing and Powell's Method." *Proc. of ASME Design Engineering Conference*, Irvine, CA, pp. 1-8.
- 38 **Vasiliu, A., and B. Yannou.** (1998). "Dimensional Synthesis of Planar Path Generator Linkages Using Neural Networks." *Mechanism and Machine Theory*, **32**(65), pp. 299-310.
- 39 **Bawab, S., et al.** (1997). "Automatic Synthesis of Crank Driven Fourbar Mechanisms for Two, Three, or Four Position Motion Generation." *Journal of Mechanical Design*, **119**(June), pp. 225-231.
- 40 **Liu, A. X., and T. L. Yang.** (1999). "Finding All Solutions to Unconstrained Nonlinear Optimization for Approximate Synthesis of Planar Linkages Using Continuation Method." *Journal of Mechanical Design*, **121**(3), pp. 368-374.
- 41 **Beyer, R.** (1963). *The Kinematic Synthesis of Mechanisms*. McGraw-Hill: New York. p. 254.
- 42 **Hall, A. S.** (1961). *Kinematics and Linkage Design*. Prentice-Hall: Englewood Cliffs, NJ, p. 49.

5.16 PROBLEMS[‡]

Note that all three-position synthesis problems below may be done using a matrix solving calculator, equation solver such as *Mathcad*, *Matlab*, or *TKSolver*, program *MATRIX*, or program *LINKAGES*. Two-position synthesis problems can be done with a four-function calculator.

- 5-1 Redo Problem 3-3 using the analytical methods of this chapter.
- 5-2 Redo Problem 3-4 using the analytical methods of this chapter.
- 5-3 Redo Problem 3-5 using the analytical methods of this chapter.
- 5-4 Redo Problem 3-6 using the analytical methods of this chapter.
- 5-5 See Project P3-8. Define three positions of the boat and analytically synthesize a linkage to move through them.
- 5-6 See Project P3-30. Define three positions of the dumpster and analytically synthesize a linkage to move through them. The fixed pivots must be located on the existing truck.
- 5-7 See Project P3-7. Define three positions of the computer monitor and analytically synthesize a linkage to move through them. The fixed pivots must be located on the floor or wall.
- *[†]5-8 Design a linkage to carry the body in Figure P5-1 through the two positions P_1 and P_2 at the angles shown in the figure. Use analytical synthesis without regard for the fixed pivots shown. Hint: Try the free choice values $z = 1.075$, $\phi = 204.4^\circ$, $\beta_2 = -27^\circ$; $s = 1.24$, $\psi = 74^\circ$, $\gamma_2 = -40^\circ$.
- [†]5-9 Design a linkage to carry the body in Figure P5-1 through the two positions P_2 and P_3 at the angles shown in the figure. Use analytical synthesis without regard for the fixed pivots shown. Hint: First try a rough graphical solution to create realistic values for free choices.
- [†]5-10 Design a linkage to carry the body in Figure P5-1 through the three positions P_1 , P_2 , and P_3 at the angles shown in the figure. Use analytical synthesis without regard for the fixed pivots shown. Hint: Try the free choice values $\beta_2 = 30^\circ$, $\beta_3 = 60^\circ$; $\gamma_2 = -10^\circ$, $\gamma_3 = 25^\circ$.
- *[†]5-11 Design a linkage to carry the body in Figure P5-1 through the three positions P_1 , P_2 , and P_3 at the angles shown in the figure. Use analytical synthesis and design it for the fixed pivots shown.
- [†]5-12 Design a linkage to carry the body in Figure P5-2 through the two positions P_1 and P_2 at the angles shown in the figure. Use analytical synthesis without regard for the fixed pivots shown. Hint: Try the free choice values $z = 2$, $\phi = 150^\circ$, $\beta_2 = 30^\circ$; $s = 3$, $\psi = -50^\circ$, $\gamma_2 = 40^\circ$.
- [†]5-13 Design a linkage to carry the body in Figure P5-2 through the two positions P_2 and P_3 at the angles shown in the figure. Use analytical synthesis without regard for the fixed pivots shown. Hint: First try a rough graphical solution to create realistic values for free choices.
- [†]5-14 Design a linkage to carry the body in Figure P5-2 through the three positions P_1 , P_2 , and P_3 at the angles shown in the figure. Use analytical synthesis without regard for the fixed pivots shown.

TABLE P5-0

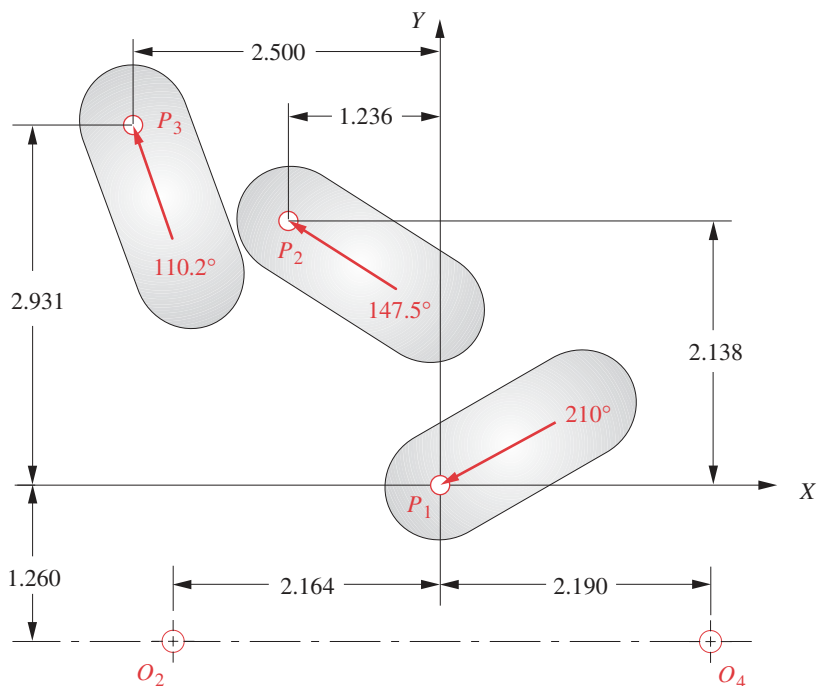
Topic/Problem Matrix

5.2 Two-Position Rocker Synthesis
5-51, 5-52, 5-53
5.4 Two-Position Motion Generation
5-1, 5-2, 5-8, 5-9, 5-12, 5-13, 5-16, 5-17, 5-21, 5-22, 5-23, 5-54, 5-55, 5-56, 5-59, 5-60
5.7 Three-Position Motion Generation
5-3, 5-10, 5-14, 5-18, 5-24, 5-25, 5-27, 5-28, 5-31, 5-32, 5-34, 5-37, 5-38, 5-39, 5-41, 5-42, 5-44, 5-45, 5-57, 5-61, 5-64
5.9 Synthesis for A Specified Fixed Pivot Location
5-4, 5-5, 5-6, 5-7, 5-11, 5-15, 5-19, 5-26, 5-29, 5-30, 5-33, 5-35, 5-36, 5-40, 5-43, 5-46, 5-62, 5-65
5.10 Center-Point and Circle-Point Circles
5-20, 5-47, 5-48, 5-49, 5-50, 5-58, 5-63

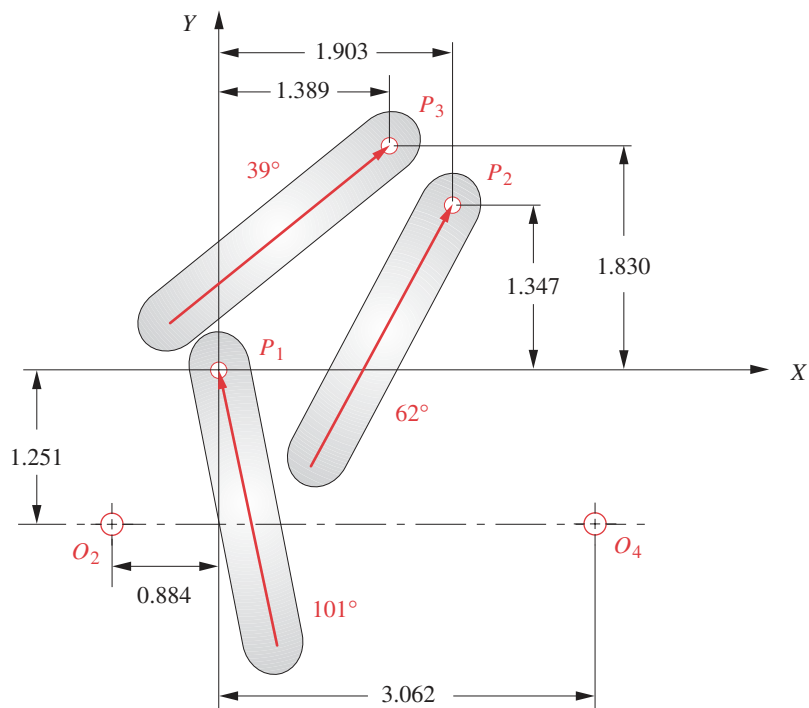
[‡] All problem figures are provided as PDF files, and some are also provided as animated AVI and Working Model files. PDF file names are the same as the figure number. Run the file *Animations.html* to access and run the animations.

* Answers in Appendix F.

[†] These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs. In most cases, your solution

**FIGURE P5-1**

Data for Problems 5-8 to 5-11

**FIGURE P5-2**

Data for Problems 5-12 to 5-15

- *†5-15 Design a linkage to carry the body in Figure P5-2 through the three positions P_1 , P_2 , and P_3 at the angles shown in the figure. Use analytical synthesis and design it for the fixed pivots shown.
- †5-16 Design a linkage to carry the body in Figure P5-3 through the two positions P_1 and P_2 at the angles shown in the figure. Use analytical synthesis without regard for the fixed pivots shown.
- †5-17 Design a linkage to carry the body in Figure P5-3 through the two positions P_2 and P_3 at the angles shown in the figure. Use analytical synthesis without regard for the fixed pivots shown.
- †5-18 Design a linkage to carry the body in Figure P5-3 through the three positions P_1 , P_2 , and P_3 at the angles shown in the figure. Use analytical synthesis without regard for the fixed pivots shown.
- *†5-19 Design a linkage to carry the body in Figure P5-3 through the three positions P_1 , P_2 , and P_3 at the angles shown in the figure. Use analytical synthesis and design it for the fixed pivots shown.
- †5-20 Write a program to generate and plot the circle-point and center-point circles for Problem 5-19 using an equation solver or any programming language.

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs. In most cases, your solution can be checked with program LINKAGES.

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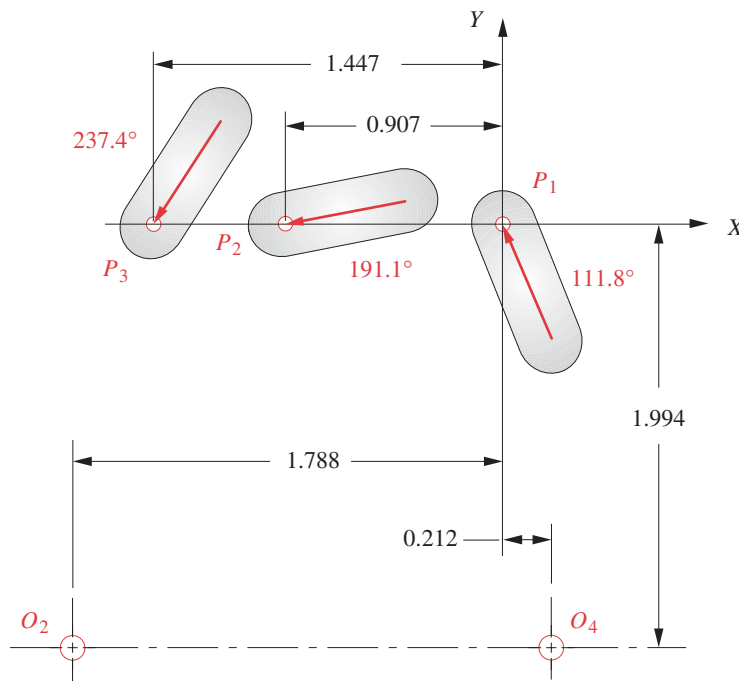


FIGURE P5-3

Data for Problems 5-16 to 5-20

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs. In most cases, your solution can be checked with program LINKAGES.

- 5



Data for Problems 5-21 to 5-26

- †5-27 Design a fourbar linkage to carry the object in Figure P5-5 through the three positions shown in their numbered order without regard for the fixed pivots shown. Use any points on the object as attachment points. The fixed pivots should be on the base. Determine the range of the transmission angle.
- †5-28 Design a fourbar linkage to carry the object in Figure P5-5 through the three positions shown in their numbered order without regard for the fixed pivots shown. Use points *A* and *B* for your attachment points. Determine the range of the transmission angle.
- †5-29 Design a fourbar linkage to carry the object in Figure P5-5 through the three positions shown in their numbered order using the fixed pivots shown. Determine the range of the transmission angle.
- †5-30 To the linkage solution from Problem 5-29, add a driver dyad with a crank to control the motion of your fourbar so that it cannot move beyond positions one and three.
- †5-31 Design a fourbar linkage to carry the object in Figure P5-6 through the three positions shown in their numbered order without regard for the fixed pivots shown. Use points *A* and *B* for your attachment points. Determine the range of the transmission angle.
- †5-32 Design a fourbar linkage to carry the object in Figure P5-6 through the three positions shown in their numbered order without regard for the fixed pivots shown. Use any points on the object as attachment points. The fixed pivots should be on the base. Determine the range of the transmission angle.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs. In most cases, your solution can be checked with program LINKAGES.

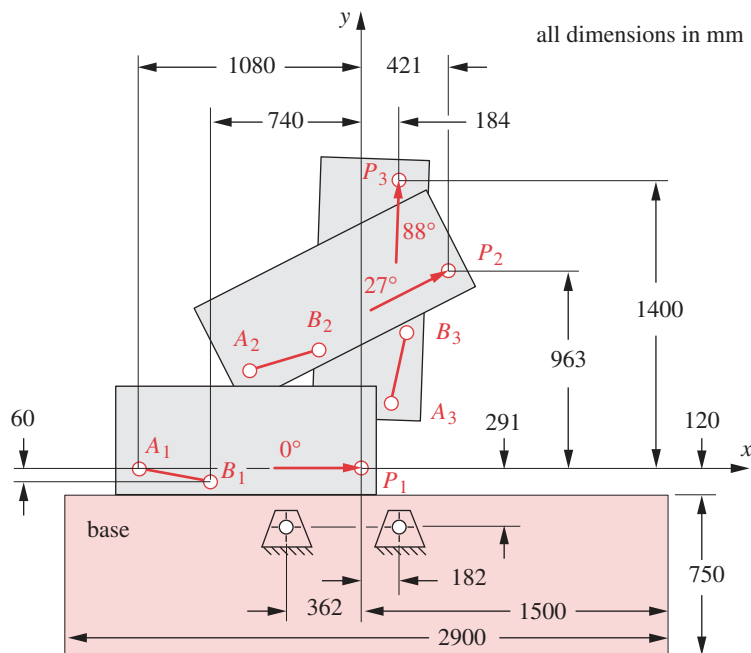


FIGURE P5-5

Data for Problems 5-27 to 5-30

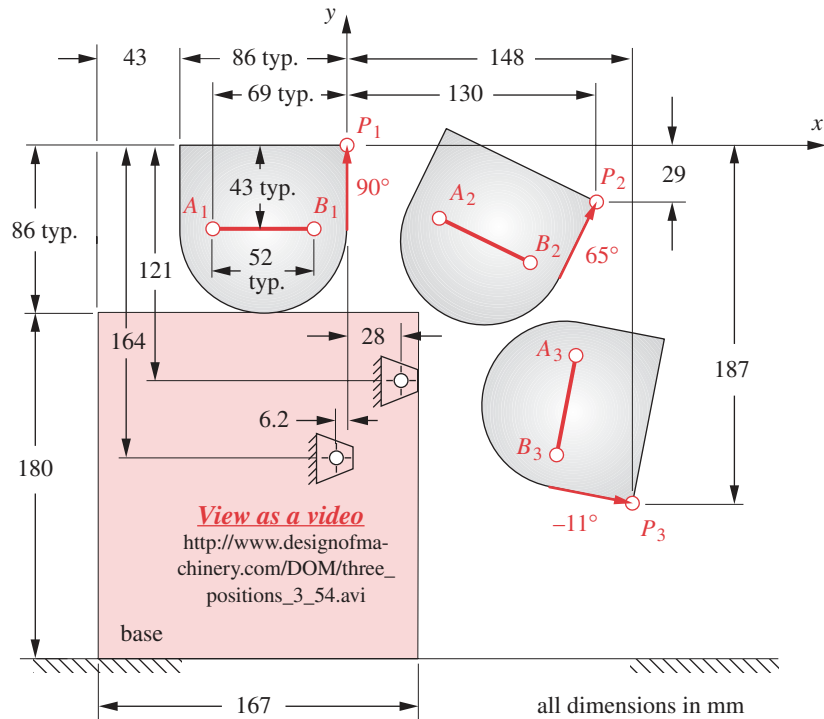


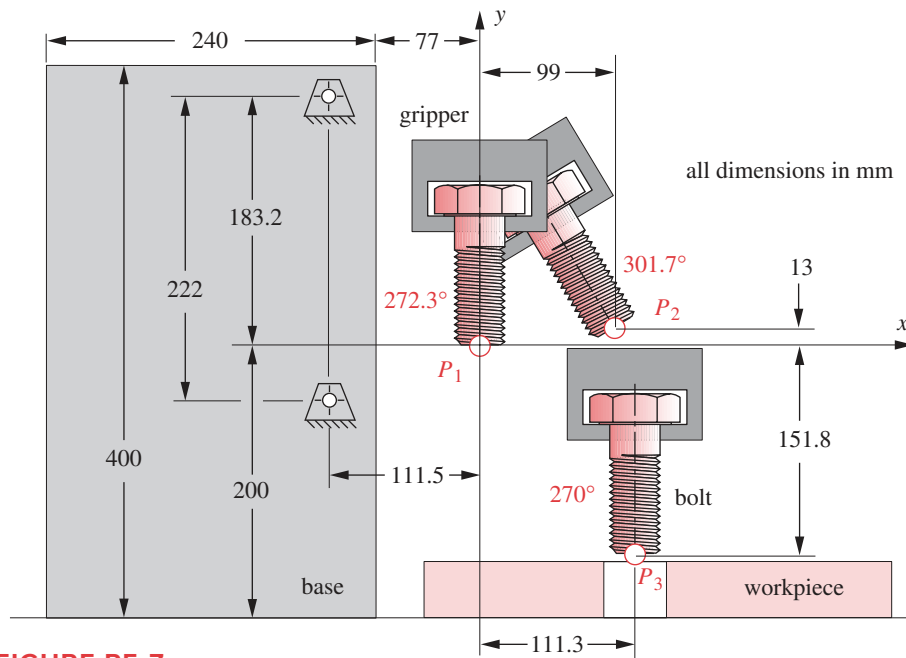
FIGURE P5-6

Data for Problems 5-31 to 5-33

* Answers in Appendix F.

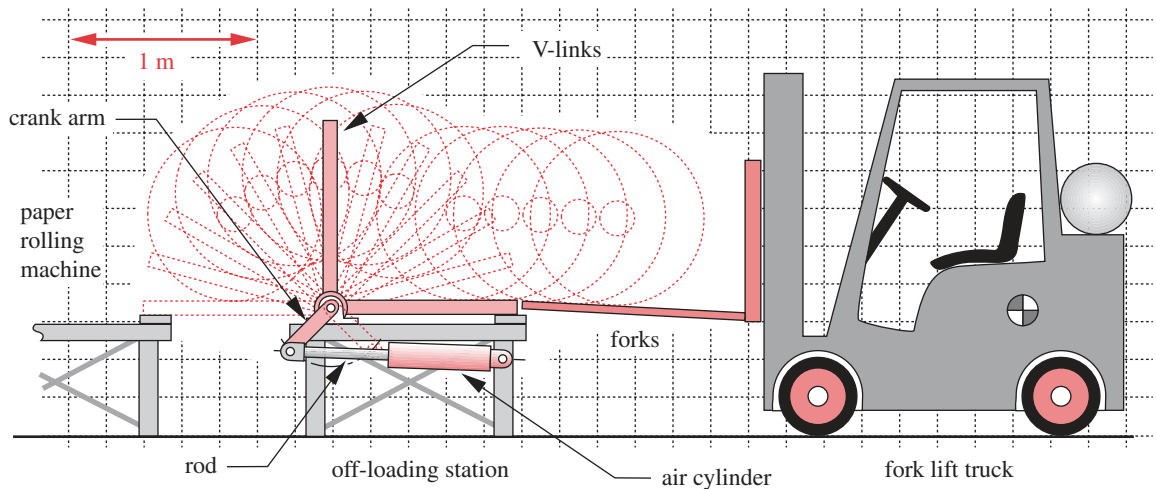
† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs. In most cases, your solution can be checked with program LINKAGES.

- *†5-33 Design a fourbar linkage to carry the object in Figure P5-6 through the three positions shown in their numbered order using the fixed pivots shown. Determine the range of the transmission angle.
- †5-34 Design a fourbar linkage to carry the bolt in Figure P5-7 from positions 1 to 2 to 3 without regard to the fixed pivots shown. The bolt is fed into the gripper in the z direction (into the paper). The gripper grabs the bolt, and your linkage moves it to position 3 to be inserted into the hole. A second degree of freedom within the gripper assembly (not shown) pushes the bolt into the hole. Extend the gripper assembly as necessary to include the moving pivots. The fixed pivots should be on the base. Hint: Try guess values of $\beta_2 = 70^\circ$, $\beta_3 = 140^\circ$, $\gamma_2 = -5^\circ$, $\gamma_3 = -49^\circ$.
- *†5-35 Design a fourbar linkage to carry the bolt in Figure P5-7 from positions 1 to 2 to 3 using the fixed pivot locations shown. Extend the gripper assembly as necessary to include the moving pivots. See Problem 5-34 for more information.
- 5-36 To the linkage solution from Problem 5-35, add a driver dyad with a crank to control the motion of your fourbar so that it cannot move beyond positions one and three.
- 5-37 Figure P5-8 shows an off-loading mechanism for paper rolls. The V-link is rotated through 90° by an air-driven fourbar slider-crank linkage. Design a pin-jointed fourbar linkage to replace the existing off-loading station and perform essentially the same

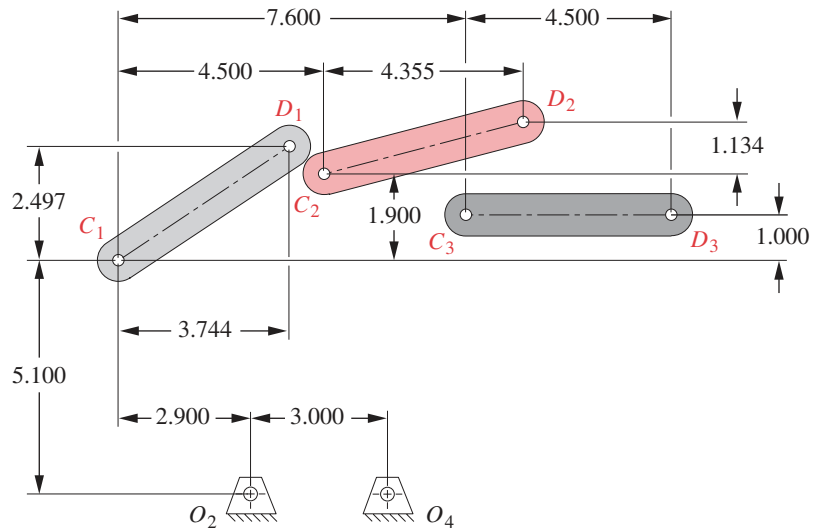
**FIGURE P5-7**

Data for Problems 5-34 to 5-36

function. Choose three positions of the roll including its two end positions and synthesize a substitute mechanism. Use a link similar to the existing V-link as one of your links. Add a driver dyad to limit its motion to the range desired.

**FIGURE P5-8**

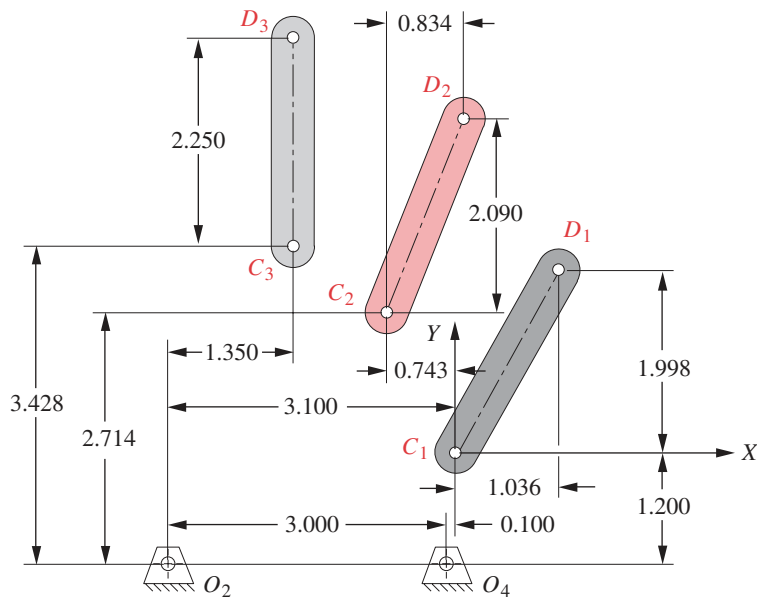
Problem 5-37

**FIGURE P5-9**

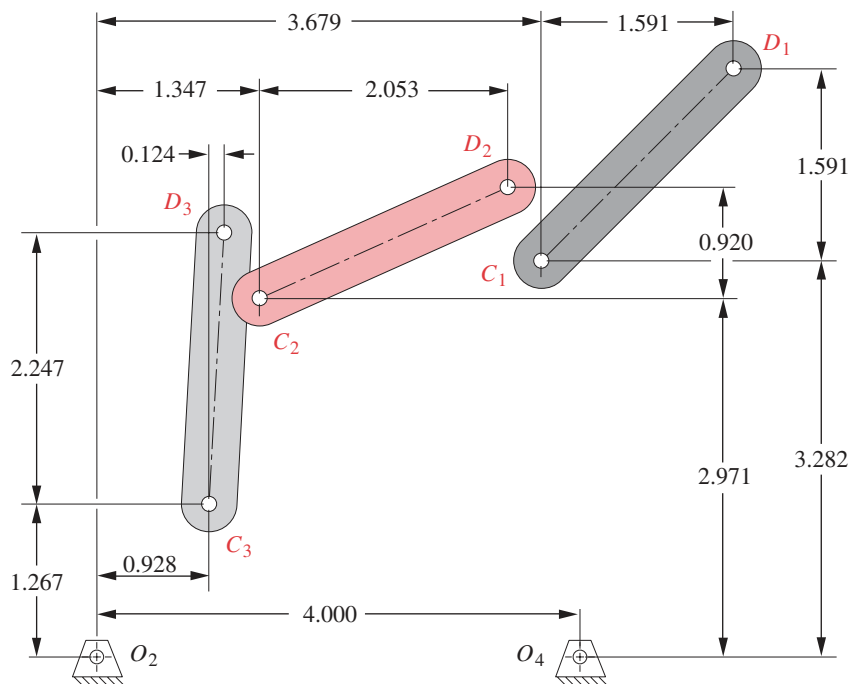
Data for problems 5-38 to 5-40 and 5-47

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs. In most cases, your solution can be checked with program LINKAGES.

- †5-38 Design a fourbar linkage to carry the object in Figure P5-9 through the three positions shown in their numbered order without regard for the fixed pivots shown. Use points C and D for your attachment points. Determine the range of the transmission angle.
- †5-39 Design a fourbar linkage to carry the object in Figure P5-9 through the three positions shown in their numbered order without regard for the fixed pivots shown. Use any points on the object as attachment points. Determine the range of the transmission angle.
- †5-40 Design a fourbar linkage to carry the object in Figure P5-9 through the three positions shown in their numbered order using the fixed pivots shown. Determine the range of the transmission angle.
- †5-41 Repeat Problem 5-38 using the data shown in Figure P5-10 instead.
- †5-42 Repeat Problem 5-39 using the data shown in Figure P5-10 instead.
- †5-43 Repeat Problem 5-40 using the data shown in Figure P5-10 instead.
- †5-44 Repeat Problem 5-38 using the data shown in Figure P5-11 instead.
- †5-45 Repeat Problem 5-39 using the data shown in Figure P5-11 instead.
- †5-46 Repeat Problem 5-40 using the data shown in Figure P5-11 instead.
- †5-47 Write a program to generate and plot the circle-point and center-point circles for Problem 5-40 using an equation solver or any programming language.
- †5-48 Repeat Problem 5-47 using the data from Problem 5-43 instead.
- †5-49 Repeat Problem 5-47 using the data from Problem 5-46 instead.

**FIGURE P5-10**

Data for problems 5-41 to 5-43 and 5-48

**FIGURE P5-11**

Data for problems 5-44 to 5-46 and 5-49

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs. In most cases, your solution can be checked with program LINKAGES.

- †5-50 In Example 5-2 the precision points and rotation angles are specified while the input and output rotation angles β and γ are free choices. Using the choices given for β_2 and γ_2 , determine the radii and center coordinates of the center-point circles for O_2 and O_4 . Plot those circles (or portions of them) and show that the choices of β_3 and γ_3 give a solution that falls on the center-point circles. You can get some verification of your circle calculations by using program LINKAGES.
- 5-51 Design a driver dyad to move link 2 of Example 5-1 from position 1 to position 2 and return.
- 5-52 Design a driver dyad to move link 2 of Example 5-2 from position 1 to position 3 and return.
- 5-53 Design a driver dyad to move link 2 of Example 5-3 from position 1 to position 3 and return.
- 5-54 Design a fourbar linkage to carry the object in Figure P5-12 from position 1 to 2 using points C and D for your attachment points. The fixed pivots should be within the indicated area.

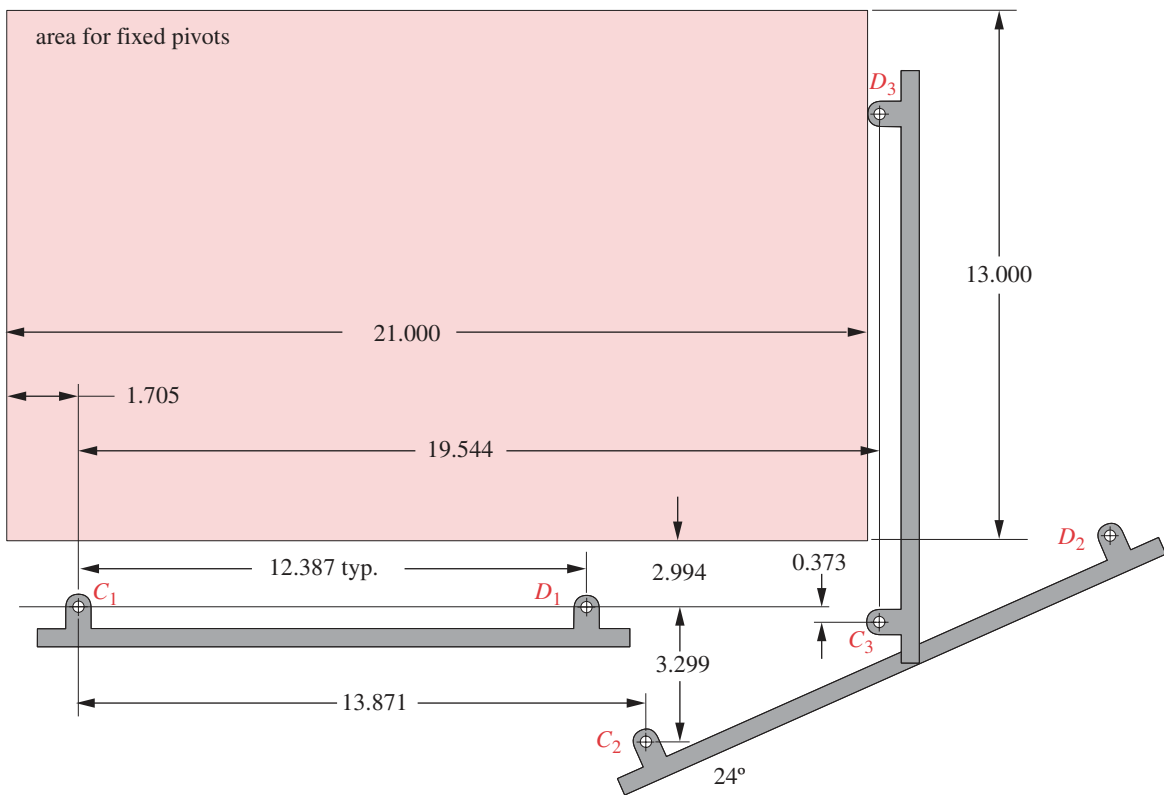
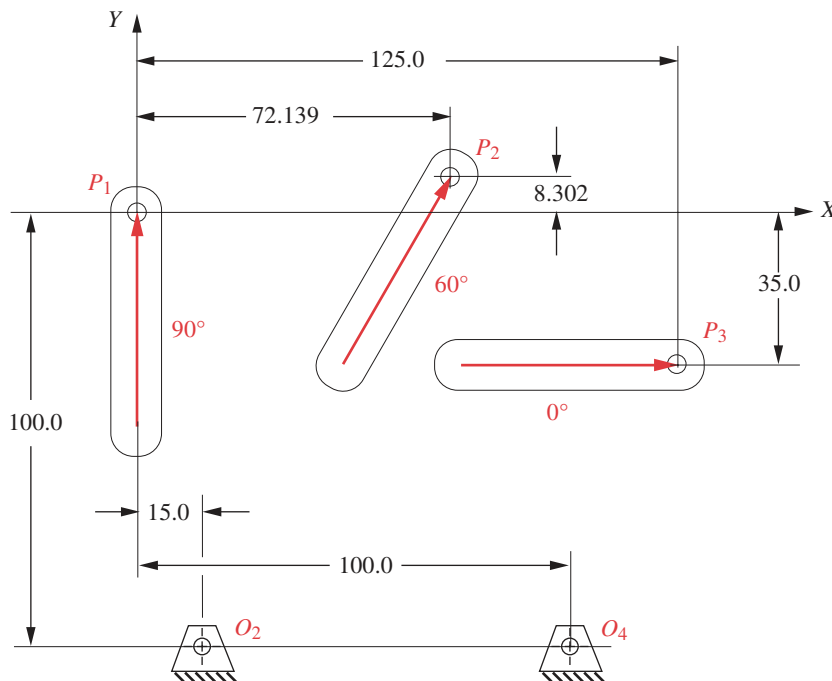


FIGURE P5-12

Problems 5-54 to 5-57

- 5-55 Design a fourbar linkage to carry the object in Figure P5-12 from position 1 to 3 using points C and D for your attachment points. The fixed pivots should be within the indicated area.
- 5-56 Design a fourbar linkage to carry the object in Figure P5-12 from position 2 to 3 using points C and D for your attachment points. The fixed pivots should be within the indicated area.
- 5-57 Design a fourbar linkage to carry the object in Figure P5-12 through the three positions shown in their numbered order using points C and D for your attachment points. The fixed pivots should be within the indicated area.
- 5-58 Write a program to generate and plot the center-point and circle point circles for Problem 5-11 using an equation solver or any program language.
- 5-59 Design a linkage to carry the body in Figure P5-13 through the two positions P_1 and P_2 at the angles shown in the figure. Use analytical synthesis without regard for the fixed pivots shown. Hint: Try the free choice values $z = 50$, $\phi = 20^\circ$, $\beta_2 = 30^\circ$, $s = 75$, $\psi = 120^\circ$, $\gamma_2 = 20^\circ$.
- 5-60 Design a linkage to carry the body in Figure P5-13 through the two positions P_2 and P_3 at the angles shown in the figure. Use analytical synthesis without regard for the fixed pivots shown. Hint: First try a rough graphical solution to create realistic values for free choices.

**FIGURE P5-13**

Problems 5-59 through 5-62

- 5-61 Design a linkage to carry the body in Figure P5-13 through the three positions P_1 , P_2 and P_3 at the angles shown in the figure. Use analytical synthesis without regard for the fixed pivots shown.
- 5-62 Design a linkage to carry the body in Figure P5-13 through the three positions P_1 , P_2 and P_3 at the angles shown in the figure. Use analytical synthesis and design it for the fixed pivots shown.
- 5-63 In Example 5-3 the precision points and rotation angles are specified while the positions of O_2 and O_4 are free choices. Using the values given for β_2 and γ_2 , determine the radii and center coordinates of the center-point circles for O_2 and O_4 . Plot those circles and show that the values of β_3 and γ_3 give a solution that falls on the center-point circles.
- 5-64 Design a linkage to carry the body in Figure P5-14 through the three positions P_1 , P_2 and P_3 at the angles shown in the figure. Use analytical synthesis without regard for the fixed pivots shown.
- 5-65 Design a linkage to carry the body in Figure P5-14 through the three positions P_1 , P_2 and P_3 at the angles shown in the figure. Use analytical synthesis and design it for the fixed pivots shown.

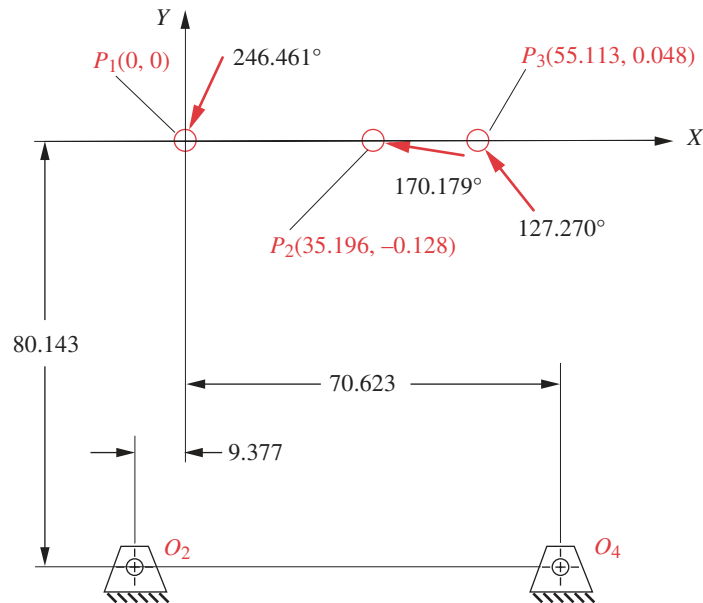


FIGURE P5-14

Problems 5-64 through 5-65